Characterizing the Student-TA Ecosystem in Computing Programs: A Two Year, Multi-University TA Training Experiment

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We present an experiment designed to evaluate the teaching assistant (TA) ecosystem in computing programs, particularly in relation to whether or not in-person, pedagogically focused TA training, based on social constructivist learning theory, impacts TA and student experiences in office hours. We collect data from computing programs at five universities over two years, amounting to more than 300 TA responses and 4,600 student responses. While previous studies in computing education have identified TAs' potential to increase student sense of belonging as well as which stances TAs adopt to demonstrate care towards students, to our knowledge, this work represents the widest spanning collection of student and TA data to date, both specifically within computing and in general. We make three important contributions to computing education research from quantitative analysis of our data. First, we identify race, course, university, class standing, and whether or not a TA has been trained as significant factors for TA experiences. Specifically, for effects of training, we find that TAs who receive *any* training are more confident in encouraging questions and engagement, creating a welcoming environment, telling students when they do not know an answer, and conducting office hours. Our second contribution characterizes student experiences in office hours. We show that student disability status is associated with significantly different office hours experiences across universities, followed by race, course, and term. Our final contribution focuses on the students who *do not* attend office hours. To our knowledge, this is the first study that analyzes students in computing programs who do not attend office hours as well as those who do attend office hours. We find that students with historically marginalized identities are more likely to report avoiding office hours due to being *externally*-avoidant, being pushed away by external factors, such as not feeling comfortable with any TAs for the course, than their peers. In contrast, students with historically privileged identities are more likely to report not attending office hours because they are *internally*-avoidant, reporting that they do not need help. Finally, we show that students indicating they have a disability report both worse experiences in office hours *and* more reticence to attend office hours, leaving them in a particularly challenging situation. These findings specifically speak to student and TA experiences within computing; this is a field that not only has a well-documented and unique history of marginalization based on identity characteristics such as race and gender, it has *also* seen a boom in enrollment since the early 2000s that has been answered, in part, by computing programs relying on TAs to provide instructional support. Given this unique setting, we create a detailed portrait of the student-TA ecosystem in computing across several universities, characterizing TA experiences, student experiences, and differences that emerge along demographic lines.

CCS Concepts: • **Social and professional topics → Computing education**.

Additional Key Words and Phrases: Teaching Assistants, Broadening Participation in Computing, TAs, office hours, student-TA interactions

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1 Introduction

Teaching assistants (TAs) form a critical link from professors to students. Because they are situated as near-peers of students, TAs have a strong potential to be deployed in support of broadening participation in computing (BPC) efforts [22, 30]. Studies across STEM disciplines have shown that both students and TAs benefit from peer learning [4, 30, 47, 62, 63, 68]. In fact, in a 2023 meta-analytic investigation, Fong et al. identified a small but meaningful correlation between self-reported help-seeking behavior and student achievement across 108 studies considering both student-teacher and student-TA interactions [24]. TAs have the potential to impact the classroom in ways instructors cannot and have unique insights that instructors may benefit from.

In this work, we use a broad definition of the term "TA"; when we say "TA", we assume that this is an undergraduate or graduate student worker whose typical role includes working directly with students in office hours, as well as grading assignments, providing administrative assistance, and possibly working in a lab/class setting [51]. Office hours themselves can range from primarily one-on-one interactions with a single TA and a single student meeting in an office or common space to a TA working with an entire room of students, moving among them to address questions as they arise. All office hours at institutions in this study are drop-in; they do not require students to sign up for appointments in advance. TAs in this group include those who are hired based on applications as well as those who receive TA appointments as part of funding packages provided by the university. In this work, we include class standing as an explicit dimension for our analyses of differences in experience because undergraduate and graduate TAs are not only at different points in their academic careers, they are also drawn from populations with different demographic distributions at the participating universities.

Turning to TAs in computing specifically, in 1995, Roberts et al. found that TAs build stronger rapport with students in large classes due to their level of direct contact with students [77]. More recent work shows that TAs may help: 1) reduce the impacts of the gaps in previous computing experience often present in introductory computing courses; 2) help students value wrong answers as useful and; 3) encourage students to view each other as collaborators [8]. Previous work has also found providing support to TAs themselves can be important for building an inclusive culture, especially in the introductory sequence where feelings of belonging may be particularly nuanced [53]. One further benefit of effectively deploying TAs is that the experience of TAing itself is a legitimate site of learning for the TAs themselves [30, 66, 85].

From a BPC perspective, previous work has found that TAs have the capacity to help build and support an inclusive culture within computing departments. Ensuring that TAs are trained using effective pedagogical techniques–those techniques that have been shown to help learners retain information–such as peer instruction [47], team teaching [66], and peer-to-peer learning [83, 88] has been posited to have a positive impact on student learning by ensuring that TAs are better prepared [9, 23, 55, 69, 92**?**]. This better preparation may manifest through higher awareness of cultivating a welcoming environment [9] and differences in student identity and experience [55] as well as by more effectively guiding activities such as pair programming [23, 69]. While studies show these as impacts TAs can have on students, no studies have been able to isolate the impacts of TA training specifically but turning to education research, several studies find that teacher certification and pedagogical content knowledge have positive effects on teacher quality and effectiveness [21, 31, 37].

One example of a project focused on TA training and mentorship [26] has led to forums that focus on creating TA training and TA professional development programs as ways to facilitate inclusive learning spaces [55, 59]. For instance, in a 2022 study, Weikle et al. focus efforts at their university on using TAs to form specific support networks

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for historically marginalized student populations [86]. Data show that for historically marginalized students feelings of belonging are nuanced, and that providing support to TAs in these areas can be important for building an inclusive culture, especially in the intro sequence in computing programs [53].

Motivated by the specific context of encouraging students to leverage TAs in computing courses in the United States, we set out to measure the effects on TAs and students of providing in-person, synchronous training to TAs. We turn specifically to the context of office hours for two reasons: 1) office hours are consistent in purpose and structure from university to university for the institutions participating in this study in spite of structural diversity in the TA programs overall and 2) previous work has shed qualitative insights that on specific benefits, such as an increased sense of belonging, that students may gain from office hours interactions [52, 68].

Within this context, we implement a TA training based in supporting positive interpersonal interaction between students and TAs with the aim of testing its efficacy. Recent work, explored more in Section 2, from Perlmutter et al. and Molina Villegas et al. notes interpersonal interaction as particularly prescient during office hours interactions [52, 68].

Because this work is motivated in part by BPC efforts, we note that there have been a growing number of calls within this space to conduct data analysis without aggregating racial and ethnic categories together [15, 54, 71, 91]. We follow these calls and present our work intersectionally where possible so that unique experiences within race and gender are not hidden. This is specifically relevant to studying the TA-student ecosystem because of the nuanced feelings of belonging that Pournaghshband and Medel identified as important to diversity-inclusive pedagogy [71]. To this end, we offer analytic contributions by evaluating our data without creating aggregate categories for race and gender when possible, allowing for further insight into dynamics that researchers have already identified, such as differences in experiences between South versus East Asian students [91].

In all, we present the results of an experiment deploying TA training at five different institutions across the U.S. We take advantage of the fact that all five institutions use TAs to host office hours to draw insight that evaluates TA experiences, TA perceptions of TA training, student experiences with TAs during office hours, and the reasons that students choose not to attend office hours.

Our experiment took place at five different institutions across the U.S. over a two year span; we focus on office hours specifically as the site of student-TA interaction because all TAs at all universities in all courses evaluated in this study hold office hours, which cannot be said for other sites of student-TA interaction. To the best of our knowledge, it represents the most extensive dataset gathered to date characterizing the student-TA ecosystem. As the first of its scope, this study has two goals: 1) to understand how to evaluate the TA ecosystem–TAs, students, and student-TA interactions during office hours and 2) to determine how to evaluate the impacts on office hours and TA self-perception of deploying TA training. More specifically, these goals lead us to five research questions, shown in Table 1.[1]

Note that it is not an assumption of this work that all students *should* go to TA office hours. Rather, it is an assumption that students should not be driven to *avoid* office hours based on their lived experiences or perceptions of the TAs.

First, we review the prior literature. Next, we describe our experimental design (Section 3), the methods we use (Section 4), the data we collect (Section 5), and the limitations of this study (Section 6). In Section 7 we turn to analysis of the 313 TA survey responses in our dataset before working with the 4,646 student responses in Section 8. Finally, we present a discussion of our results and conclusions (Section 9), and future work (Section 10).

[1]We release our survey tools to the community in Appendix B with extended results tables in Appendix C.

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	Question	Results	Sections
TA-RQ1	What factors contribute to TA comfort, confidence, and perception of their skills?	race, course, class standing	Section 7.1
TA-RQ2	What is TA perception of the training they receive? What factors influence this perception?	positive; race, class standing	Section 7.2
TA-RQ3	Does TA perception of their abilities and experiences change when they receive in-person, pedagogy-focused training, versus logistics-only, asynchronous, or no training?	when TAs receive no training at all, they are less confident; differentiation between types of training was not detected in this study	Section 7.2
S-RQ1	Are students going to office hours? To what extent is this due to *internal* factors (e.g., "I did not need help.") versus *external* (e.g., "I did not feel comfortable with any of the TAs for this course.") factors?	the majority of students do not attend office hours; the majority of those who do not attend believe they do not need help, but reasons are not uniformly distributed according to demographic factors and historically marginalized in computing are more likely to be *externally*-avoidant	Section 8.2
S-RQ2	What factors contribute to student experience in office hours?	disability status, race, course	Section 8.1

Table 1. Research questions studied, separated by primary population concerned (**TA**s or **S**tudents); a summary of results are displayed alongside notes on where in the paper they are addressed for reader convenience

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2 Related Work

In our survey of related work, we focus on the benefits of deploying TA programs and evaluation of a) the training programs and of TAs themselves b) student-TA interactions and c) TA office hours, specifically in the context of computing education. While each subtopic is addressed separately, together, they work to define 1) why computing programs use TAs, 2) why TA training is needed and how it has been evaluated thus far, and 3) the specific context of office hours.

2.1 Benefits of Deploying TAs

Research shows that the weed-out culture pervasive in engineering is a formidable barrier to retaining students from historically marginalized groups [19]. Not only are successful student-TA interactions associated with higher achievement [24, 89], TAs are situated in a key position to re-write such narratives and push back on the idea that introductory computing courses are only "for" students who have prior programming experience [8]. For instance, Perlmutter et al. identify three key areas in which TAs can increase student sense of belonging: competence, relatedness, and autonomy [68]. Of course, just as these interactions have the potential to be affirming interactions, they have the potential to be discouraging ones. For example, when CS 1[2] students report factors that affect their decisions about whether or not to seek help, several factors are negative, such as judgment and intimidation, alongside the more positive familiarity and trust [38, 39].

[2]CS 1 refers in this paper to the first course required in the CS degree. It is typically an introduction to programming course.

2.2 TA Training Programs

As noted earlier, there is a long line of research within education examining the effects of teacher preparation programs in a K-12 context. In this context, several studies find that teacher certification and pedagogical content knowledge have positive effects on teacher quality and effectiveness [21, 31, 37].

Transitioning to the domain of higher education within STEM as a whole, there are a number of examples of deploying and training student assistants, e.g., in Chemistry and Physics [63, 84]. Otero et al. describes the Learning Assistant (LA) program,[3] which has a robust and mature training program in which LAs enroll in a for-credit training course that itself also incorporates experienced LAs. Finally, turning to computing-specific TA training programs, early work such as Kay's 1995 contribution proposed a seminar series throughout the term for first-time TAs in CS [32]. Since then, there have been a number of partial and complete descriptions of deployed TA training programs, most of which are less intensive than a full-term course and vary in modality [2, 55, 70, 82, 87, 92].

The body of research on TAs in computing as a tends to focus on the structure of programs in general rather than TA training specifically [48, 51, 73]. In their in-depth review of forty existent papers studying undergraduate TAs (UTAs) in computing, Mirza et al. found that only twenty of these papers mentioned TA training. Of these twenty papers, sixteen describe training course content in some way, but this ranges from just a few sentences to being the focus of the paper. Indeed, Mirza et al. specifically highlight that in 2019, the development of practices related to training that promotes diversity is "not well studied or reported" [51]. In 2023, Perlmutter et al. gave an overarching recommendation to use TAs to increase student belonging by providing TA training that emphasizes understanding student context and the diversity of student experiences and identities [68].

2.3 Evaluating TA Training Programs

A small number of studies have been conducted to evaluate TA training programs in computing. Danielsiek et al. evaluate a TA training program[4] that is largely focused on logistics at several large German universities (working with 56 TAs total); they find that the largest gains for TAs are in greater understanding of their roles as TAs [20]. Both Minnes' and Riese and Kann's 2022 studies use blended learning techniques with the TAs they train. Both conclude that all training makes TAs feel more prepared. Figures within these studies show that TAs prefer synchronous formats versus online, asynchronous modules [50, 75]. In 2025, Zaman et al. described a TA training program using asynchronous modules, but also note that engaging TAs is difficult and that this decision was made based on resource constraints [93], concluding that though not an ideal format, their deployed TA training program is beneficial to TAs and, implicitly, to students based on their results from 34 TA participants.

2.4 TA Self-Evaluation

Benefits for the students working as TAs have been identified by past work, which finds boosts in self-efficacy in both computing [30] and in STEM more broadly [4]. Some case studies have examined TAs' experiences from their own perspectives. In a study with eight interviewed TAs and 23 observed lab sections, Patitsas identifies the attributes of labs that lead to higher TA satisfaction, such as room layout, lighting, and length of lab sessions [65]. A less easily quantifiable factor, having a positive social environment, was also identified as an important support.

[3] https://www.learningassistantalliance.org/
[4] https://www.uni-muenster.de/Ketti/en/

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Another project that analyzes a TA from her own perspective is that of Markel and Guo [46]. These authors conduct analysis of "firsthand accounts from an undergraduate TA documenting her 36 hours of in-lab tutoring for a CS 2[5] course, where she engaged in 69 one-on-one help sessions." However, the purpose of this study was to draw insights to develop useful scaffolding for TAs and form a model of student-TA interaction, rather than to evaluate the TA experience specifically.

In a large-scale evaluation of challenges that TAs face, Riese et al. conduct a thematic analysis of 180 reflective essays from TAs at three European institutions, identifying professionalism, student interactions, assessment, and using best teaching practice as areas TAs struggle in across all three institutions [76].

Finally, in a 2025 case study, 10 UTAs were interviewed at the University of California, San Diego [52]. This 2025 study focuses on understanding how TAs view their roles around student learning and providing support. The authors specifically aim to identify what might lead to"sub-optimal TA behavior" such as long queues of students and pressure to give students solutions. We are not aware of any studies that have attempted to quantify TAs' evaluations of their own experiences at scale, either at a single university or across universities.

2.5 Evaluating TAs as Teachers

In terms of evaluating TAs as teachers, Patitsas and Belleville as well as Santiesteban et al. focus on identifying correlations between 1) student evaluations of TAs and student performance and 2) student evaluations of TAs and attributes of the TAs, such as race and gender [67, 80]. Patitsas and Belleville studied three years of records from CS 1 and two years of records from CS 2 and an introductory course in discrete math at the University of British Columbia. In this context, TAs teach labs in addition to grading and holding office hours. They found that in CS 1, students benefit from working with UTAs more than with graduate TAs [67]. Unlike for professors [36], differences in student evaluations of TAs based on gender were not found by either of these studies. Along the dimensions of experiences with TAs, final grades, and student retention, Patitsas and Belleville found only that poor TA interactions lead to lower rates of student retention, which adds to similar finding from O'Neal et al. at an American university [61].

Some evaluation of TAs as teachers appears as a by product of describing a new strategy in managing or working with TAs, such as Leyzberg et al.'s description of a new strategy for interviewing and selecting TAs in a large introductory course [42]. The authors then evaluate the interview process via student evaluations of TAs tracked over two years. Though this work displays results in support of improved student-TA interactions with a growth from 50% to 60% of TAs being evaluated as "excellent", the results are not evaluated for statistical significance or multiple courses.

2.6 Evaluating student-TA interactions

Three studies have presented qualitative work on understanding student-TA interactions from the student perspective since 2023. Perlmutter et al. reveal the possibilities of increasing student sense of belonging through student-TA interactions [68]. Also in 2023, Lim et al. interviewed 15 students to learn about student expectations and desires when attending tutoring sessions, identifying points of friction between student desires and student learning [43]. Barkhuff et al.'s 2025 study is the only one of these three to evaluate student-TA interactions beyond a single institution, extending their analysis to two institutions. These authors ultimately identify that TAs adopt humanistic stances to demonstrate care towards students [9].

[5]CS 2 refers to the second course required in the CS degree. It is typically an intermediate programming course.

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2.7 Evaluating TA Office Hours

Though there has been some work conducted on TA office hours in computing, the kind of help students seek [40, 74], and why they attend office hours [34], to the best of our knowledge, there has not been prior work in the specific context of computing on why students *do not* attend office hours. Across six courses in an engineering program (not including computer engineering), Robinson et al. find three groups of students with low office hours attendance: those feeling they understand the material, those who procrastinate, and those who study less overall [78]. All existent follow-up work to this study focuses on STEM broadly [28, 29] or engineering specifically [1].

2.8 Intersectional Analytic Framework

For this study, where possible, we conduct our analysis through an intersectional lens. Specifically, we focus on categories of difference as a framework to understand that within racial categories, gender may impact the way people experience the world [3]. This framework can be used to guide methodology for gaining insight into secondary marginalization–groups that are historically marginalized along more than one dimension, i.e., Southeast Asian women–and analyzing whether or not observed effects are amplified when multiple dimensions of marginalization are isolated. Within computing, over the last two decades, there has been an increasing focus on evaluating statistics intersectionally, that is, along dimensions of both race and gender simultaneously [54, 60, 71] to ensure that the experiences of any particular subgroup of the population is not hidden [27]. Indeed, in the key conclusions and recommendations of the the 2021 National Academies report on "Transforming Trajectories for Women of Color in Tech" [56] the authors write

> ...the lack of disaggregated data poses a major challenge to understanding the nuanced and specific needs of different subgroups of women of color.

Research has shown that Black women in computing have a consistently different experience than Black men and non-Black women [72, 79], that Asian students and Asian women in particular have different experiences than their peers [91], and that Latinx students have their own unique intersectional experiences [44].

Usage of this framework means that in our analysis, interpretation begins from an explicit understanding that population-level judgments, though useful in indicating possible areas for intervention and future study, will not fully capture the unique experiences of individuals.

2.9 Summary

In all, evaluating and describing TA programs, TA training, and student-TA interactions is complex–especially when large course sizes are involved–making it difficult to guarantee student interaction with a specific set of TAs. More often than not, studies of the TAs and the student-TA ecosystem are small (under 20 participants) and at single institutions, bringing a wealth of important qualitative insight. We seek to contribute a broad, landscape view of this ecosystem through quantitative evaluation of TA training and office hours across institutions, conducting our analysis through an intersectional lens when possible.

3 Experimental Design

In Section 1, we described the goals of this work as two-fold: first, to understand how to evaluate the TA ecosystem–TAs, students, and student-TA interactions–and second, to determine how to evaluate the effects of deploying a specific TA training. To this end, we ask seek to answer five specific research questions, displayed in Table 1. The three research questions that concern TAs focus on understanding their perceptions of their abilities and the impacts of TA training.

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The two research questions that concern students focus on their experiences in office hours and the impacts of TA training on these experiences.

In this experiment, we aim to test the impact of deploying in-person, pedagogy-focused TA training in a natural environment. We take advantage of the fact that many universities currently deploy TAs with little-to-no training. When training exists, prior work and the author's experiences with multiple institutions find that it often focuses only on logistical aspects of the role and may be online and asynchronous. Existing training often fails to use or teach effective teaching techniques such as team teaching [66] and peer-to-peer learning [83, 88], among other topics. In the author's experience, when in-depth training is provided, it often depends on one specific person and stops when that person leaves or changes roles.

	Univ. 1	Univ. 2	Univ. 3	Univ. 4	Univ. 5
R1/R2?	R1	R2	R1	R1	R1
# CS 1 students	400 - 500	100 - 150	400 - 500	300 - 600	400 - 500
# CS 1 TAs	20 - 25	3 - 7	20 - 30	10 - 20	20 - 25
GTAs?	✓	✓	no	✓	✓
UTAs?	✓	✓	✓	no	✓
Baseline term	FA2023	SP2023	FA2023	SP2023	FA2023
Previous training	Logistics only	Online/ Async.	Logistics only*	Logistics only	Online/ Async.
Target TA pop.	CS 1, 2, 3	CS 0, 1, 2	New TAs	New TAs	New TAs
Intervention format	1 x 2hrs	1 full day	2 x 2hrs	4 hrs	2 x 2hrs
Grew TA training team?	no	no	no	yes	no
Continuing TA training in 2025?	yes	yes	no**	yes	yes

Table 2. Characterizations of universities and their intervention TA training programs; * = staff reported that a TA training had happened in the past but the content of the training was unknown and it was not sustained; ** = The faculty running the training left the institution after Fall 2024

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First, the authors obtained approval from their institution's Institutional Review Board (IRB) to conduct this study. This approval was granted based on the experimental design laid out below, the contents of surveys deployed (shown in Section 3.1 and Appendix B), the methods of obtaining participant consent (shown in Appendix A), the optional nature of both taking the surveys and of answering all questions, and the anonymity of the survey. No data presented in this study was collected from students who knew the authors or attended the author's institution. Furthermore, all data presented to the community and to participant institutions is aggregated across at least one dimension, if not more. The IRB offices of all universities involved in this study were contacted and letters of support were obtained.

Universities were verbally recruited for this experiment by the author during online learning workshops about TA training curricula. Inclusion criteria were that the university must have TAs assigned to introductory CS courses who work directly with students in office hours. Exclusion criteria were: 1) already having a synchronous TA training program not primarily focused on policies and regulations, 2) not having at least 10 TAs identified to participate in the training, or 3) if a university only deploys student employees who do not interact with students via office hours (e.g., graders). Of the eight universities that began the experiment, five maintained the TA training programs developed for this experiment and gathered data for the duration of the study, conducting in-person, synchronous, pedagogically-focused training at the beginning of each term and deploying surveys to students and TAs at the end. These five universities are all public universities located across the U.S. and are further characterized by the attributes below in Table 2.

- **R1/R2**: Carnegie classification indicating the research level of the institution in the U.S.; R1 indicates very high research activity, R2 high research activity
- **CS 1 students**: indicates how large a typical CS 1 course is at the institution across all sections
- **CS 1 TAs**: indicates how many TAs a typical CS 1 course employs at the institution across all sections
- **GTAs**: whether or not the university deploys graduate TAs (GTAs)
- **UTAs**: whether or not the university deploys undergraduate TAs (UTAs)
- **Baseline term**: the first term data was collected, *before* the intervention TA training was deployed
- **Target TA population**: which TAs attend the deployed TA training
- **Intervention format**: how long the deployed TA training is in number of sessions and hours
- **Grew TA training team**: whether or not the institution expanded the team of teachers that lead the TA training
- **Continuing TA training in 2025**: whether or not the TA training continued past the scope of this experiment

For each university, we started by identifying the target populations among students and TAs. All universities collected data from students and TAs in their CS 1 and CS 2 courses. Determinations about which courses were identified as CS 0, 1, 2, and 3[6] were made by the universities themselves. Some universities collected data from additional first- and second-year courses. Some universities also collected data from all first-term TAs.

Next, we worked with each university to design an in-person, synchronous TA training centered on interpersonal communication and pedagogy. Materials were provided to each university for several training modules, including ones addressing common student-TA interactions, contextualizing the TA job and the values held by students, TAs, and professors, pedagogical strategies for engaging with students in office hours, and the effects identity may have in both classroom and office hours settings through an intersectional lens as recommended by previous literature [15, 54, 71, 91]. Each module includes activities and guidelines for structuring discussion based on the number of TAs in the training and the sensitivity of the content in the module. The majority of content involves small group discussions that are then shared with the larger group, guided by the instructor. This training does not focus on teaching formal pedagogical frameworks though it does introduce concepts such as the revised Bloom's taxonomy of learning [35] and open versus closed questions [64]. Depending on the size of the target population and the institutional constraints the university was working with–such as the number of instructors and compensation available–we adjusted the curriculum as necessary. Training at each university ranged from 2 hours (Univ. 1) to one full day (Univ. 2). All trainings drew from the same underlying curriculum, described and released in full by Muzny and Shah in 2023 [55], a set of training modules based on a social constructivist framework of learning [90] in which TAs work together to formulate appropriate responses to situations and answers to prompts couched in the particular social context of computing in U.S. universities.

Each university collected data at the end of a baseline term, then launched TA training the following term with the help of the authors of this study. Two universities began the experiment in Spring 2023 and three began in Fall 2023. The experiment concluded at the end of Fall 2024. All TAs and students were surveyed at the end of each term. Surveys were sent to the respective populations by the faculty teaching the corresponding courses at each university. We recommended but could not enforce surveys be distributed during class and TA meetings.[7]

The conclusions that we draw are limited by the experimental design specifically in two ways. First, we only collected one term (versus an academic year) of baseline data. Second, we do not know which students interacted with trained

[6]CS 0 a non-majors course or an optional course before CS 1, CS 2 is the second course, CS 3 is the typical third course.
[7]To our knowledge, one university, Univ. 5, gave a small amount of credit for survey completion, and Univ. 3 distributed surveys in class. Univ. 4 distributed surveys asynchronously (email and/or course discussion board). Survey distribution mechanisms for Univ. 1 and Univ. 2 were not reported. Univ. 5 maintained anonymity by giving students credit based on submitting a screenshot of the survey completion screen.

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TAs; none of our institutions had the capacity to train all TAs every term and every institution had open office hours, with students able to see any available TA. These limitations are discussed in Section 6.

3.1 Surveys

We deployed two surveys at the end of each term: one for students and one for TAs.

The student survey was designed to evaluate student interactions with TAs throughout the term and was written by the Computing Research Association's Center for Evaluating Research Pipeline (CERP) [6]. The questions in this survey are shown in Tables 3, 4, and 5 alongside mean scores. They include 10 general statements about students' interactions with TAs, three overall questions about office hours, and four selectable reasons for not attending office hours. All questions were rated on a Likert scale from 1 to 5, with corresponding wording provided by CERP.[8] Individual students may have responded to this survey at the end of different terms and courses. Because TAs are not shared between courses, we consider these responses to be distinct from each other as students are reporting on experiences with a unique set of TAs for a unique course for that particular term.

Num.	Question	μ	SD	N	TA-CQ
S-SQ1	The TAs were knowledgeable about the course content.	4.0	0.9	1909	TA-CQ1
S-SQ2	The TAs explained course concepts clearly.	3.9	1.0	1904	TA-CQ2
S-SQ3	The TAs encouraged my questions and engagement.	3.9	1.1	1906	TA-CQ3
S-SQ4	The TAs created a welcoming environment.	4.2	1.0	1886	TA-CQ4
S-SQ5	The TAs gave clear explanations to my questions.	3.9	1.0	1887	
S-SQ6	The TAs were respectful to me.	4.4	0.9	1884	
S-SQ7	I felt comfortable asking for help from the TAs.	4.1	1.0	1888	
S-SQ8	The TA office hours were easy to access.	3.9	1.1	1868	
S-SQ9	I felt that the TAs wanted me to succeed.	4.1	1.0	1868	
S-SQ10	I would feel comfortable asking the TAs for advice about academic matters not related to this course.	3.5	1.3	1880	

Table 3. S-SQ (**S**tudent-**S**ub **Q**uestion) questions with mean scores across all data; TA-CQ column indicates the corresponding TA question; 1 = "not at all true", 5 = "very true"

Num.	Question	μ	SD	N
S-OQ1	How would you describe your overall experience(s) with TAs during office hours for this course?	4.3	0.9	1870
S-OQ2	Overall, how successful were TAs in creating an environment that was conducive to learning during office hours for this course?	4.0	1.0	1867
S-OQ3	How would you rate the overall effectiveness of the TAs' instruction and support during office hours for this course?	4.0	1.0	1868

Table 4. S-OQ (**S**tudent-**O**verall **Q**uestion) questions and mean scores across all data; S-OQ1 & S-OQ2 1 = "very negative", 5 = "very positive"; S-OQ3 1 = "not at all effective", 5 = "very effective"

We then created a TA survey by first writing corollary questions to those in the student survey, as shown in Tables 3 and 6. For example, students were asked if TAs explained course concepts clearly whereas TAs were asked how confident they are in their ability to explain course concepts clearly. Two questions were added to this set based on the

[8]Validity and reliability metrics were not provided to the authors.

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Num.	Reason	%
S-NOH1	I did not need help.	65.7%
S-NOH2	The office hour times were inconvenient for my schedule.	34.1%
S-NOH3	I did not feel comfortable with any of the TAs for this course.	6.1%
S-NOH4	I did not believe the TAs would be able to help me with my needs for this course.	8.1%

Table 5. Reasons for not attending office hours (**S**tudent-**N**o **O**ffice **H**ours), shown with the overall percent of students who did not go to office hours who selected each reason; reasons are not exclusive; N = 2,644

author's experiences working with TAs and the specific research questions of this study: 1) comfort telling someone the TA doesn't know the answer to a question and 2) general comfort conducting office hours.

Num.	Question	μ	SD	S-SQ
TA-CQ1	I am knowledgeable about the course content for which I am a TA.	4.5	0.8	S-SQ1
TA-CQ2	I can explain course concepts clearly for the course for which I am a TA.	4.4	0.8	S-SQ2
TA-CQ3	I know how to encourage questions and engagement.	4.2	0.8	S-SQ3
TA-CQ4	I know how to create a welcoming environment in office hours.	4.4	0.8	S-SQ4
TA-CQ5	I am comfortable telling someone that I don't know the answer to a question.	4.3	0.8	
TA-CQ6	I am comfortable conducting office hours.	4.5	0.8	
Average		4.4	0.6	

Table 6. TA-CQ (**TA**-**C**omfort **Q**uestion) questions and mean scores across all data; S-SQ column indicates the corresponding student question; 1 = "not at all true", 5 = "very true"; N = 281

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Finally, we wrote new questions pertaining to TA training specifically. These questions were taken from a survey that had been previously given at the author's institution to TAs who had gone through their own TA training programs. These questions were written by a TA training team, which included faculty members teaching all levels of courses. The Likert scale wording for these questions was taken from the original CERP survey. As with students, individual TAs may have responded to this survey at the ends of different terms. Because the TA experience with students, training, and the general TA-ecosystem changes from term-to-term, we again consider submissions from the same TA but different terms to be distinct from each other as they reflect the unique experience of that TA during that term.

Num.	Question	μ	SD
TA-TQ1	I found TA training helpful.	3.9	1.0
TA-TQ2	TA training made me more confident performing my duties as a TA.	3.7	1.2
TA-TQ3	TA training made me more comfortable performing my duties as a TA.	3.7	1.2
TA-TQ4	TA training changed how I approach students in the course I am TAing.	3.5	1.2
TA-TQ5	TA training changed how I communicate with the professor I am working with.	3.3	1.4
Average		3.6	1.1
TA-AmQ	How much TA training would you like? (1 = "significantly less", 5 = "significantly more")	3.1	0.8

Table 7. TA-TQ (**TA**-**T**raining **Q**uestion) questions and TA-AmQ (**TA**-**Am**ount **Q**uestion) with mean scores across all data; TA-TQ1 - TA-TQ5 1 = "not at all effective", 5 = "very effective"; N = 209, 272 for TA-AmQ, which includes TAs who did not receive training

We focus our analysis on all TA questions (Tables 6 and 7) and S-OQs and S-NOHs (Tables 4 and 5) for students. [9]

[9]We omit the S-SQs from our in-depth analysis because an ordinary least squares regression analysis confirms that all eleven appear as significant factors for at least one of the three S-OQs. Therefore, we conclude that no S-SQ will contain significant information that is not captured in the three S-OQs.

4 Methods

When analyzing the survey response data, we use the same techniques with all responses. For all questions, the null hypothesis is that all groups' answers to the question come from the same underlying distribution. Phrased in terms of our intervention, the null hypothesis assumes that student and TA experiences remain the same regardless of the training TAs receive. Rejecting the null hypothesis means that there is a significant difference in these distributions–indicating that, for example, TAs who receive training are more likely to give higher scores to the target question.

First, we limit comparison groups to those with at least 10 data points to ensure validity and generalizability of tests. This is also a mechanism to protect those students and TAs who have identities less represented in this dataset. [10] For TAs, because of the overall smaller number of responses, we further restrict this requirement to groups who have at least 10 responses that are spread across at least three universities.

Next, we test our distributions for normality using measurements of skew and kurtosis [17, 18]. The majority of our survey responses do not follow normal distributions, but are, rather, centered on the 4th point of a five point Likert scale.[11] For tests comparing two groups' responses, we primarily use the Mann-Whitney U test; when answers form normal distributions, we use Welch's T-test. Finally, when we compare more than two groups' responses, we use ANOVA testing to determine if the groups are likely drawn from different underlying distributions [41].[12]

Because we are performing many hypothesis tests, we must control for false positives to avoid spuriously rejecting the null hypothesis. We use the Benjamini-Hochberg false discovery rate controlling procedure to correct all p-values for multiple testing [10]. To calculate effect sizes, we use rank-biserial correlation coefficients when dealing with Mann-Whitney U tests[13] and eta squared (η^2) values with ANOVA tests[14] Finally, to avoid combinatorial explosion, we limit the tests of our data to the following factors:

- **University**: Univ. 1; Univ. 2; Univ. 3; Univ. 4; Univ. 5
- **Term**: SP2023; FA2023; SP2024; FA2024
- **Course**: CS 0; CS 1; CS 2; CS 3; Discrete 1 (Disc. 1); Other (Oth.)
- **Gender**: Man (M); Woman (W); Other (Oth.);[15] Unknown (Unk.)[16]
- **Race**: African American/African/Black (Black); Arab/Middle Eastern (Arab/ME); Caucasian/European/White (White); East Asian (E. Asian);[17] Indigenous or First Nations (Ind./FN); Native American/Alaska Native (AIAN); Native Hawaiian/Pacific Islander (HPI); Other Asian (Oth. Asian); Other (Oth.); South Asian (S. Asian);[18] Southeast Asian (SE. Asian);[19] Multi-racial (Multi); Unknown (Unk.)
- **Ethnicity**: Another Hispanic, Latinx, or Spanish origin (Other Hisp.); Cuban; Mexican, Mexican American, Chicanx (Chicanx); Not of Hispanic, Latinx, or Spanish origin (Not H/L/S); Puerto Rican; Multi; Unknown (Unk.)
- **Disability Status**: Whether or not the respondent indicated that they had a disability: Yes; None; Unknown [20]

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[10] For example, we are unable to make any comparisons for AIAN/Ind. students because no university has 10 or more respondents with this identity (Table 10).
[11] It is a known phenomenon for people to be reticent to give neutral or negative reviews, leading to high average scores [94].
[12] ANOVA has been shown to be robust for small sample sizes, data that is not normally distributed, and Likert-scale data [58, 81].
[13] Bounds used are: under 0.1 = insignificant, 0.1 - 0.3 = small, 0.3 - 0.5 = medium, over 0.5 = large [11]; no significant effects found for questions with normal distributions.
[14] Bounds used are: under 0.01 = insignificant, 0.01 - 0.06 = small, 0.06 - 0.14 = medium, over 0.14 = large [11].
[15] Respondents who indicated non-cisgender and/or non-man/woman gender identities are binned into "Other".
[16] Unknowns appear in our dataset when respondents answer the content questions but do not answer some or all of the demographic questions; these responses are omitted from demographic analysis as appropriate.
[17] China, Hong Kong, Japan, Macau, Mongolia, North Korea, South Korea, and Taiwan [16].
[18] Afghanistan, Bangladesh, Bhutan, India, the Maldives, Nepal, Pakistan, and Sri Lanka [14].
[19] Brunei, Cambodia, East Timor, Indonesia, Laos, Malaysia, Myanmar, Philippines, Singapore, Thailand, Vietnam [12].
[20] Unfortunately, our data is too sparse to disaggregate by specific type of disability to be statistically tested. While treating this group as a monolith is not ideal, the lack of previous exploration of disability status at all means that even in aggregate this analysis provides valuable new insight. Among students

- **TA Training - TAs only**: Whether the respondent had 1) in-person TA training 2) asynchronous online training or 3) no training. TAs may indicate that they received both options 1 and 2: In Person; Async; None
- **Experience - TAs only**: Whether or not a TA is a first term TA
- **Class Standing - TAs only**: Undergraduate (UG); Masters (MS); Ph.D.[21]

We limit intersectional tests of our data to only race and gender. While the number of intersectional tests we are able to conduct due to sparsity of responses is relatively low, we are able to make intersectional comparisons for both TAs and students, and indicate areas for future study. We include this analysis because computing education has recognized how important this analysis is [15, 54, 71, 91]; it should be performed, even when deep insights don't come to light.

When performing the majority of our student response analysis, we control for each university independently because of both the large quantity of student data collected and the wide variation in number of responses per university. The significantly smaller size of our TA dataset makes controlling for university intractable. These datasets are discussed in more detail in Section 5.

5 Data

In all, we collected 313 completed end-of-term TA surveys, enumerated by university and course in Table 8. Response rates vary greatly from course to course (e.g., 100% of CS 1 TAs to 21.4% of CS 2 TAs in FA2024 at Univ. 2), university to university, and term to term. The 4,646 completed student surveys make up the majority of our dataset. Just as with the TA data, Table 8 shows that student response rates varied dramatically from course to course, university to university, and term to term. From our meeting notes, we notice that this likely has a significant amount to do with *who* was teaching each course term to term, a phenomenon that we discuss further in Section 6. Indeed, there are only six (university, course) combinations with student response rates greater than 10% for every term studied at that university.

		TAs							Students						
Univ.	Term	CS 0	CS 1	CS 2	CS 3	Disc. 1	Oth.	#	CS 0	CS 1	CS 2	CS 3	Disc. 1	Oth.	#
1	FA23		10 (43.5)	1 (5.6)	5 (33.3)		1	17		312 (65.7)	226 (59.9)		4		542
	SP24		1 (4.3)	5 (20.8)	4 (44.4)		3	13		6 (1.4)	378 (99.0)	41 (15.1)	5		430
	FA24		5 (18.5)	7 (38.9)	6 (66.7)		2	20		2 (0.3)	272 (70.6)	198 (91.2)	6		478
2	SP23		3 (60.0)	9			4	16	14	9 (7.1)	38 (18.1)				61
	FA23	3 (33.3)	7	3 (42.9)				13	37 (14.6)	29 (17.7)	47 (25.4)				113
	SP24	1 (50.0)		1 (14.3)			1	3	27 (34.2)	7 (5.0)	57 (28.5)				91
	FA24	2 (66.7)	12 (100)	3 (21.4)			11	28	135 (72.6)	33 (21.4)	93 (47.4)				261
3	FA23	9 (81.8)	18 (69.2)	7 (50.0)		10 (71.4)	2	46	90 (54.2)	224 (45.1)	141 (57.3)	43 (18.1)	46 (20.2)	71	615
	SP24	1	20 (76.9)			7 (41.2)		28		127 (30.2)	132 (37.5)	18 (10.5)	78 (29.5)	4	359
	FA24		7 (38.9)		4 (36.4)	3 (30.0)		14		75 (19.5)		82 (39.0)	39 (21.5)		196
4	SP23		7 (50.0)	3 (42.9)	2 (50.0)	1 (8.3)	3	16	21	11 (3.3)	28 (18.1)	1 (0.8)	53 (44.2)	1	115
	FA23	1	4 (18.2)		1 (11.1)	3 (37.5)	26	35		53 (9.2)		1 (0.5)	40 (14.9)	3	97
	SP24		2 (14.3)	4 (26.7)		3 (50.0)	1	10		12 (3.0)	3 (0.7)		4 (1.9)		19
	FA24		2 (33.3)	1 (20.0)			8	11		6 (1.1)				37	43
5	FA23		4 (66.7)	5 (55.6)	1 (12.5)		18	28		209 (92.9)	21 (7.2)	17 (6.5)			247
	SP24			5 (55.6)				5		221 (98.2)	145 (50.0)	2 (0.8)			368
	FA24		2 (100)	5 (100)	1 (16.7)		2	10		106 (100)	274 (86.7)	231 (70.2)			611
Totals		17	104	59	24	27	82	313	324	1442	1855	634	275	116	4646

Table 8. TA and student response rates with number of responses and (rate); some courses do not have enrollment data

Tables 9 and 10 show the breakdown of all TA and student survey responses by race and gender.

indicating a single disability, the disability identified most frequently was ADHD (5% of all students, 53% of students indicating one disability) while the second most common was a general mental health disability, dropping to 2% of all students. 4% of all students indicated multiple disabilities.

[21]We keep MS and Ph.D. students separate because at U.S. universities MS students are often self-funded and not promised TA positions, Ph.D. students are often required to TA at least one term and sometimes must TA more than that to secure their funding.

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TAs		Afr/AfrAm/Bl		Arab/ME		White		E. Asian		S. Asian		SE. Asian		Multi		Unk.	
Univ.	Gend.	#	%	#	%	#	%	#	%	#	%	#	%	#	%	#	%
1	M	3	6.0			1	2.0			20	40.0					1	2.0
	Unk.													1	2.0	6	12.0
	W	1	2.0			3	6.0	2	4.0	11	22.0			1	2.0		
2	M					1	1.7	3	5.0	10	16.7	6	10.0	1	1.7	1	1.7
	Unk.									2	3.3					23	38.3
	W							4	6.7	7	11.7	1	1.7				
3	M			1	1.1	18	20.5	5	5.7	7	8.0	4	4.5	3	3.4		
	Unk.					3	3.4									9	10.2
	W	2	2.3	1	1.1	15	17.0	2	2.3	3	3.4			8	9.1	2	2.3
4	M	1	1.4	7	9.7	3	4.2	5	6.9	21	29.2			1	1.4	3	4.2
	Unk.					1	1.4									13	18.1
	W			1	1.4	1	1.4	4	5.6	7	9.7			1	1.4		
5	M	4	9.3	4	9.3	8	18.6	1	2.3	8	18.6	1	2.3	1	2.3	1	2.3
	Unk.					1	2.3									1	2.3
	W	1	2.3	2	4.7	1	2.3			4	9.3			1	2.3	2	4.7

Table 9. TA response demographic distribution by university, gender, and race; percentages indicate percent of student or TA responses from the corresponding university; blank cells represent identities with no corresponding survey data; Ind/FN, AIAN, HPI, Oth. Asian, and Oth. are excluded–fewer than 10 TAs across all universities selected this identifier

Students		Afr/AfrAm/Bl		Arab/ME		White		E. Asian		Ind./FN		AIAN		Oth. Asian		Oth.		S. Asian		SE. Asian		Multi		Unk.	
Univ.	Gend.	#	%	#	%	#	%	#	%	#	%	#	%	#	%	#	%	#	%	#	%	#	%	#	%
1	M	222	15.3	16	1.1	100	6.9	114	7.9	1	0.1	2	0.1	7	0.5	18	1.2	251	17.3	94	6.5	60	4.1	25	1.7
	Oth.	2	0.1	3	0.2	11	0.8	5	0.3	1	0.1					1	0.1	3	0.2	1	0.1	1	0.1		
	Unk.	14	1.0			7	0.5	6	0.4					1	0.1	2	0.1	14	1.0	3	0.2	6	0.4	54	3.7
	W	118	8.1	4	0.3	38	2.6	38	2.6	4	0.3	2	0.1	1	0.1	9	0.6	114	7.9	40	2.8	24	1.7	10	0.7
2	M	7	1.3	5	1.0	30	5.7	53	10.1					2	0.4	7	1.3	90	17.1	63	12.0	27	5.1	16	3.0
	Oth.	1	0.2			3	0.6	2	0.4	1	0.2									1	0.2	5	1.0		
	Unk.	2	0.4			2	0.4	2	0.4	1	0.2					1	0.2	6	1.1	5	1.0	1	0.2	40	7.6
	W	2	0.4	3	0.6	20	3.8	28	5.3	1	0.2			3	0.6	4	0.8	54	10.3	21	4.0	15	2.9	2	0.4
3	M	28	2.4	18	1.5	371	31.7	53	4.5	2	0.2	10	0.9	13	1.1	34	2.9	54	4.6	29	2.5	69	5.9	38	3.2
	Oth.	2	0.2			10	0.9	3	0.3					1	0.1	4	0.3	1	0.1			6	0.5	4	0.3
	Unk.	3	0.3	2	0.2	21	1.8	6	0.5							2	0.2	2	0.2			2	0.2	84	7.2
	W	15	1.3	8	0.7	150	12.8	31	2.6	2	0.2	1	0.1	5	0.4	14	1.2	12	1.0	13	1.1	34	2.9	9	0.8
4	M	17	6.2	3	1.1	83	30.3	7	2.6			2	0.7	2	0.7	2	0.7	31	11.3	6	2.2	12	4.4	6	2.2
	Oth.	1	0.4			3	1.1	1	0.4											1	0.4	1	0.4		
	Unk.			2	0.7	5	1.8											1	0.4			4	1.5	26	9.5
	W	10	3.6	1	0.4	22	8.0	3	1.1					1	0.4	2	0.7	10	3.6	1	0.4	7	2.6	1	0.4
5	M	57	4.6	39	3.2	288	23.5	24	2.0			5	0.4	31	2.5	23	1.9	190	15.5	91	7.4	60	4.9	27	2.2
	Oth.	2	0.2			12	1.0	1	0.1							2	0.2	2	0.2					1	0.1
	Unk.	1	0.1			14	1.1	2	0.2	1	0.1	1	0.1					1	0.1	7	0.6	3	0.2	40	3.3
	W	24	2.0	9	0.7	68	5.5	17	1.4					9	0.7	16	1.3	83	6.8	40	3.3	27	2.2	8	0.7

Table 10. Student response demographic distribution by university, gender, and race; percentages indicate percent of student responses from the corresponding university; blank cells represent identities with no corresponding survey data; HPI is excluded–fewer than 10 students across all universities selected this identifier

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6 Threats to Validity

As a result of both experimental design and realities of deploying a longitudinal study across diverse institutions with differing resources, there are important threats to validity to recognize before we report results.

One term of baseline data: Our baseline data covers one term per institution–either Spring 2023 or Fall 2023. There is evidence that fall/spring effects may be at play, as shown in prior work analyzing student outcome data [54]. Because of this, there are more potential confounding factors when we make term-to-term comparisons.

Lack of institutional support and resources: Of the three universities that dropped out of our study, one launched but was unable to maintain the TA training program for longer than one term due to lack of compensation/recognition

by departmental leadership. Additionally, when considering TA training program sustainability, only one of our five universities had the necessary institutional support to expand the TA training team beyond one person. Indeed, lack of resources is the reason that the institutions we worked with were unable to provide training to all TAs, focusing instead on just the inexperienced ones as the site for largest impact given limited resources.[22]

Difficulty of long-term studies: This study spanned two years. Even at institutions with full support, the enthusiasm to help with data collection waned after the initial launch of TA training in the first intervention term.

Within-institution uneven survey deployment: We found that the best response-rates were given when surveys were deployed in-person during class (for students) or during meetings (for TAs). Response rates were further boosted when instructors offered very small amounts of credit to students for completing the survey. Because instructor assignments to each individual course changed term to term and the instructors were ultimately responsible for deploying the surveys, certain courses had wildly different response rates, shown in Table 8.

Lack of guaranteed student-TA interaction: Because all TAs for all introductory courses were not trained every term due to limited personnel capacity at each individual university, we cannot guarantee that each student response maps to interactions with trained TAs. In fact, because of how many office hours are designed in large, introductory courses in the U.S., to know exactly who each student saw would require surveying them repeatedly each time they interact with any TA because the TAs are not assigned to specific students.

In Section 10, we discuss some modifications that may be made in the future to reduce or avoid these threats.

7 Teaching Assistant Results

Observing the matched student and TA questions in Tables 3 and 6, we see an immediate result: TAs are more confident in their abilities than students are in their corresponding experiences with TAs. For these questions students' mean score is an average of 0.4 points lower than TAs.

To directly answer TA-RQ1 and TA-RQ2, we ask TAs two corresponding groups of questions: a set about their confidence and comfort performing various TA duties (TA-CQs, Table 6) and a set about their perception of TA training (TA-TQs, Table 7). TA-RQ3, addressing whether or not in-person, pedagogy-focused training affected TAs, is answered from both these sets of questions by comparing the answers from TAs who received in-person, pedagogy-focused training versus those who did not. Finally, we ask TAs about what amount of training they feel they would prefer (TA-AmQ), ranging from significantly less to significantly more training.

All TA survey analysis is aggregated across universities to protect those TAs with identities that are uncommon in this dataset. We are therefore able to include responses from lower-representation groups, such as SE. Asian TAs.

Several factors frequently influence reported TA experience. Race, course, university, class standing, and training type are all indicated as significant on three or more of the TA comfort questions. For example, we find that undergraduate TAs report significantly higher confidence in their knowledge than TAs who are Ph.D. students. Next, we explore each dimension where significant differences occur. We begin by first analyzing the confidence and comfort questions in Section 7.1 before turning to TA training perceptions in Section 7.2. Appendix C shows full results in addition to those highlighted in the following analysis.

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[22]Anecdotally, this is an extremely common state of affairs in the U.S., especially at large institutions.

7.1 TA Confidence and Comfort (TA-CQs)

In response to TA-RQ3, whether or not TA perception of their own abilities changes when they receive in-person, pedagogy-focused training, we find evidence that TAs are more comfortable and confident with their duties when they receive training. While we do not find evidence that our specific intervention training improved TA confidence in term-to-term comparisons, these results do show that having *any* TA training is positive. We also find race, course, university, and class standing to play significant roles in TAs' experiences.

	N	TA-CQ 1	TA-CQ 2	TA-CQ 3	TA-CQ 4	TA-CQ 5	TA-CQ 6	Average
received training	211	4.5	4.4	4.2* (+++)	4.4* (+++)	4.4* (+++)	4.5* (+++)	4.4* (+++)
no training	102	4.4	4.4	4.0* (+++)	4.1* (+++)	4.1* (+++)	4.3* (+++)	4.2* (+++)

Table 11. Mean TA response values by training status; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$ all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses for significant differences, - indicates insignificant, + indicates small, ++ indicates medium, +++ indicates large

7.1.1 Impacts of Training Type. TA-CQ3, TA-CQ4, TA-CQ5, and TA-CQ6, highlighted in Table 11, are flagged as significantly different with large effect sizes when TAs are provided with no training versus when they are provided with training. These questions correspond to TA perception of their abilities to encourage questions and engagement, create a welcoming environment, comfort level telling students they do not know an answer, and comfort conducting office hours in general.

TAs who received training are more comfortable with encouraging questions and engagement, creating a welcoming environment, telling someone they do not know an answer, and holding office hours in general than TAs who did not. While our hypothesis was that TAs who specifically received in-person training would see benefits over TAs who did not, our data does not support this conclusion; by the end of the term, any impacts from differences in TA training have normalized. Indeed, tests for the intervention training versus previous, asynchronous, trainings do not show differences.

7.1.2 Other Significant Factors. To address TA-RQ1 beyond the specific dimension of training, we describe the differences in reported confidence for TA skills along race, course, and class standing below.[23]

	N	TA-CQ1	TA-CQ2	TA-CQ3	TA-CQ4	TA-CQ5	TA-CQ6	Average
Black	12	4.2*** (+++)	4.1*** (++)	3.8** (++)	4.6	4.6* (+)	4.7* (+)	4.3** (++)
Arab/ME	16	3.8*** (+++)	3.8*** (++)	3.8** (++)	4.3	4.6* (+)	4.5* (+)	4.1** (++)
White	57	4.7*** (+++)	4.6*** (++)	4.2** (++)	4.3	4.2* (+)	4.5* (+)	4.4** (++)
E. Asian	28	4.3*** (+++)	4.3*** (++)	4.0** (++)	4.2	4.0* (+)	4.4* (+)	4.2** (++)
S. Asian	100	4.6*** (+++)	4.6*** (++)	4.4** (++)	4.5	4.3* (+)	4.7* (+)	4.5** (++)
SE. Asian	13	4.7*** (+++)	4.4*** (++)	3.9** (++)	3.9	4.8* (+)	4.1* (+)	4.3** (++)
Multi	18	4.1*** (+++)	4.1*** (++)	3.9** (++)	4.2	4.3* (+)	4.2* (+)	4.1** (++)

Table 12. TA-CQs: Mean TA response values by race; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$ all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses for significant differences, - indicates insignificant, + indicates small, ++ indicates medium, +++ indicates large

[23]While university was significant, there were no identifiable patterns. Significant differences were not detected along the other dimensions studied.

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Race: Averages for the significantly different questions along racial lines generally follow the same pattern: for the first three questions, S. Asian and white TAs are among the top three groups, with SE. Asian joining for TA-CQ1 and TA-CQ2. S. Asian TAs in particular consistently rate their skills, confidence, comfort, and the benefits of training among the highest in relation to their peers. Black and Arab/ME TAs are in the top three groups for both TA-CQ 5, which asks about level of comfort telling students [the TAs] do not know the answer to a question, and TA-CQ 6, which asks about comfort conducting office hours. Counterintuitively, both of these racial groups were in the bottom three for the questions that asked about confidence with specific aspects of TA-student interactions. Phrased another way, two of the groups that report being most comfortable with office hours are also the least confident in their content knowledge, explanations, and engagement with the students.

Race and Gender: We find that in this dataset, race appears to dominate demographic influence on TA comfort and experience, however, due to sparsity in answers, we were only able to test responses from E. Asian, S. Asian, and White TAs intersectionally with gender. We found no differences for these groups.

Course: Significant differences in reported TA experience along lines of course arose for two of the TA-CQs. CS 1 TAs have the highest reported confidence in their content knowledge (4.7 versus 4.5 for CS 0, CS 2, and CS 3, 4.3 for Discrete 1, and 4.1 for other courses) while confidence in ability to explain course concepts fell in the order CS 3 (4.7), CS 1 (4.6), CS 2 (4.5), CS 0 (4.4), Discrete 1 (4.2) and other courses (4.0).

Class Standing (UG, MS, Ph.D.): We find three TA-CQs where TAs with different class standing have statistically significantly different responses–though with a small effect size–the questions about confidence in creating a welcoming environment, in their own knowledge, and in their ability to explain course concepts clearly. Surprisingly, Ph.D. students rate their confidence the lowest on average among these groups (4.2 versus 4.5 (MS) and 4.6 (UG) for confidence in knowledge and 4.2 versus 4.4 (MS) and 4.5 (UG) for ability to explain course concepts clearly). We theorize that this could be due to several factors: how long the student has been at the institution, whether or not TAing is a graduation requirement, likelihood of speaking English as a native language, length of time since taking the course, and mismatch between the equivalent course at the TA's previous institution. We leave the investigation of these questions for future work.

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7.2 TA Perception of Training

To address TA-RQ2, TA perception of the training they receive, we analyze our target factors for differences in response to questions TA-TQ1 through TA-TQ5. With an average rating for helpfulness of TA training of 3.9, we can say that TAs find training to be helpful. However, we do uncover significant differences in evaluation of TA training based on TA race, course, and class standing.

Two dimensions were noticeable in their absence in the list of significant factors: 1) in-person and asynchronous training types and 2) whether or not a TA is in their first term. This result indicates that by the end of the term, TAs view the training that they receive similarly, regardless of which training it is and of whether or not they were a first-term TA (220 responses) or an experienced TA (93 responses). We discuss further implications and caveats of this evidence in Section 9, including what we believe may be effects of when these surveys were deployed.

7.2.1 Impacts of Training Type. By the end of the term, TAs perceive the effects of TA training the same, no matter which training they receive. Additionally, by the end of the term, there is no difference in how much more or less training TAs desire based on which training they received or even if they did not receive training at all. This is in spite of the findings from the previous section revealing that TAs are, in fact, more comfortable when they do receive

training. All TAs indicate that they are relatively happy with the amount of training they received, with a slight skew towards wanting more training, as indicated in Table 7.

7.2.2 Other Significant Factors. Race, class standing, and course were indicated to be significant in regard to TA experience of training. While analysis of the other factors, including intersectional race and gender, was conducted, others were not indicated as significant.[24]

	N	TA-TQ1	TA-TQ2	TA-TQ3	TA-TQ4	TA-TQ5	TA-AmQ
Black	12	4.0*** (++)	4.0** (++)	3.9*** (++)	3.5** (++)	3.5** (++)	3.4** (++)
Arab/ME	16	3.7*** (++)	3.6** (++)	3.8*** (++)	3.4** (++)	3.2** (++)	2.6** (++)
White	57	3.5*** (++)	3.4** (++)	3.3*** (++)	3.1** (++)	2.9** (++)	3.1** (++)
E. Asian	28	3.6*** (++)	3.6** (++)	3.5*** (++)	3.4** (++)	3.2** (++)	3.1** (++)
S. Asian	100	4.2*** (++)	4.1** (++)	4.2*** (++)	3.9** (++)	3.7** (++)	3.2** (++)
SE. Asian	13	3.2*** (++)	3.1** (++)	3.0*** (++)	3.1** (++)	3.4** (++)	3.2** (++)
Multi	18	3.4*** (++)	3.4** (++)	3.3*** (++)	3.2** (++)	2.7** (++)	3.5** (++)

Table 13. Mean TA response values by race; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$ all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses for significant differences, - indicates insignificant, + indicates small, ++ indicates medium, +++ indicates large

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Race: For all TA training questions, S. Asian TAs are associated with the highest mean rating. Interestingly, S. Asian TAs are slightly above average in their response to TA-AmQ–meaning that they want *more* training than the overall average even though they were also among the top reporting groups for their comfort and confidence inspected in Section 7.1. White, SE. Asian, and Multi-racial TAs universally rate the helpfulness and effects of TA training the lowest. While SE. Asian and Multi-racial TAs are among the groups that most want more TA training, white TAs have the second least desire for more training. We leave study of potential cultural influences in these results to future study.
Class Standing: Though there is not a significant difference in how much more or less training TAs want when dividing by class standing, there is significant difference for all questions regarding perceived helpfulness and effects of training. MS students find the training to be most helpful (4.2), reporting TA training to have increased their confidence, comfort, and changed how they interacted with both students and professors. UG students rated the helpfulness lowest at 3.6 with Ph.D. students averaging 3.8. Our observations about MS TAs correspond with the patterns we saw in Section 7.1–S. Asian students are more represented among MS TAs[25] and also report the highest average scores.
Course: We see fairly weak signals (at $p < 0.1$) for differences along lines of course for all TA-TQs except for TA-TQ1, whether or not the TAs found training helpful. With small to medium effect sizes, TAs for CS 2 report the most impacts from TA training on their TA experience as well as the most desire for more TA training.

8 Student Results

Of the 4,646 student survey responses we gathered, 2,002 of them were from students who attended office hours (43.1%), leaving the majority of respondents never having attended office hours. Table 14 shows office hours attendance rates by university and term, highlighting that the rate at which students attend office hours varies greatly university to university. So long as students are attending office hours when they need help, low attendance rates in and of themselves are not problematic. Section 8.2 analyzes the reasons students do not attend office hours.

[24]Table 21 shows all values and notes that intersectional tests could only be conducted for White, E. Asian, and S. Asian TAs.
[25]68.3% of MS TAs are S. Asian versus 14.4% of undergraduate and 31.9% of Ph.D. TAs.

term university	SP2023	FA2023	SP2024	FA2024
Univ. 1		32.5	27.0	24.1
Univ. 2	32.8	22.1	23.1	15.7
Univ. 3		62.9	69.4	65.8
Univ. 4	35.7	45.4	31.6	39.5
Univ. 5		45.7	46.7	54.0

Table 14. Percent of students surveyed who attended office hours by university and term

At two of our five universities, women attend office hours at higher rates than their male peers, though with small to insignificant effect sizes. [26] At two of our universities, Black students attend at higher rates than their peers; at two universities students in CS 1 attended office hours at higher rates than their peers; at one university this was inverted–students in CS 2 and CS 3 attended more office hours than in CS 1. In all, no patterns for office hours attendance emerged from the data gathered.

8.1 Student Perception of Office Hours

Overall, students responding to our survey reported positive experiences with means of 4.3, 4.0, and 4.0 for the questions about overall experience, learning environment, and instruction effectiveness. The surveys collected are influenced by selection bias–many institutions did not require students to complete the survey, though we note that the majority of our survey responses do come from courses where response rates are much higher than 10%. However, the distributions of student ratings are not bimodal for any question–if only students who had very negative or very positive experiences responded, we would expect peaks at both the lowest and highest ends of the scale. An ANOVA test further confirms the differences in student response between the three overall questions are significant at a level of $p < 0.01$–while students tend to describe their experiences very positively for all S-OQs (**S**tudent-**O**verall **Q**uestion), when asked to report specifically on the environment created (S-OQ2) and the effectiveness of the TAs' instruction (S-OQ3), the responses are significantly lower.

To answer S-RQ2, what factors contribute to student experience in office hours, we run hypothesis testing on the complete set of 2,002 survey responses corresponding to students who went to office hours at least once. We find that as a group, student surveys have significant differences in response patterns along the lines of race, disability status, course, and university. Most consistently, we see that students indicating that they have a disability versus those who indicate they do not have different experiences in office hours, as shown in Table 15. Indeed, at one of the two institutions where results were not statistically significantly different, Univ. 2, there were no students who attended office hours indicating they had a disability–only students indicating no disability. We conduct the remainder of our student analysis controlling for university. Full results tables are available in Appendix C.

We note that while it would be ideal to test the impacts of TA training on student experience in office hours, because of the limitations noted in Section 6, in particular that no university had the resources to train all TAs and that no university assigned students to specific TAs, we are not able to determine which students interacted with trained TAs.

[26] Previous studies in post-secondary education find women do attend office hours at higher rates than their male peers [24]. Note that this does not mean that a greater *number* of women attend office hours–because the number of women in computing courses at our universities is dramatically lower than men, TAs can still expect to *see* more men in office hours even when women attend at a much higher rate.

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8.1.1 Factors Influencing Student Experience in Office Hours. The clearest results from this experiment address S-RQ2–what factors influence student experience in office-hours? Disability status was the singular dimension along which we observed a clear pattern–students who indicate having a disability rate their experiences lower than students who indicate not having a disability, shown in Table 15.

question	factor	value	Univ. 1 mean	Univ. 1 N	Univ. 2 mean	Univ. 2 N	Univ. 3 mean	Univ. 3 N	Univ. 4 mean	Univ. 4 N	Univ. 5 mean	Univ. 5 N
S-OQ1	Dis.	None	4.2^{**} (+++)	320	4.4	72	4.3^{***} (+++)	499	4.4^{***} (+++)	60	4.4	489
(overall experience)		Yes	3.8^{**} (+++)	40			3.9^{***} (+++)	119	3.5^{***} (+++)	23	4.1	64
S-OQ2	Dis.	None	4.0^{**} (+++)	318	4.2	72	4.0^{***} (+++)	498	4.0^{**} (+++)	60	4.2	489
(learning environment)		Yes	3.5^{**} (+++)	40			3.7^{***} (+++)	119	3.3^{**} (+++)	23	4.0	64
s-OQ3	Dis.	None	3.9^{**} (+++)	319	4.1	72	4.0^{***} (+++)	498	4.0^{**} (+++)	60	4.2	489
(instruction effectiveness)		Yes	3.6^{**} (+++)	40			3.7^{***} (+++)	119	3.2^{**} (+++)	23	3.9	64

Table 15. Mean student scores by disability status for overall office hours questions; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; + indicates small effect size, ++ medium, +++ large

Disability status: Students with disabilities did attend office hours at four universities (Univs. 1, 3, 4, and 5); all overall office hours experience questions were flagged at three of these universities with high levels of confidence and large effect sizes. For all questions, students who indicated having a disability rated their experiences lower than those who indicated they did not have a disability (with an average difference of 0.5 points).

Race: Univ. 4 had significant differences with medium to large effect sizes along race for all three overall questions but was only able to test four racial groups against each other[27]. At universities 1, 3, and 5, where many groups were tested, significant differences with small effect size appeared for S-OQ1 and S-OQ3; no consistent patterns arose.

Course: One university, Univ. 3, saw significant differences with small effect sizes between courses; CS 3 scored lowest and CS 0 scored highest (an average difference of 0.6 points across S-OQs).

Term: Two universities, Univs. 3 and 4 saw significant differences term-to-term, but each institution only had three terms of data, making fall-to-fall and spring-to-spring comparisons intractable. One university saw increases from the baseline term to the following terms, one saw a decrease.

8.2 Students who did not attend office hours

For our final exploration of S-RQ1, whether or not students are going to office hours and why, we investigate the reasons that students do not attend office hours–56.9% of the student responses in our dataset. Our analysis shows that there are distinctive differences in rates that students indicate the reasons for not attending office hours based on demographic attributes. Within every reason offered, we uncover a stark pattern–students who are historically marginalized are more likely to select reasons indicating that they are externally-avoidant at higher rates than their peers.

We define a student being *externally*-avoidant as being pushed away by factors external to the student, such as TA office hours settings, TA behavior, and logistics of office hours. This is in contrast to *internally*-avoidant, not attending office hours because of personal, internal, factors, such as the student's own self-evaluation. This differentiates reasons like "I did not need any help" (internal) from "I did not feel comfortable with any of the TAs for this course" (external). These designations are shown with each reason in Table 14.

[27] Black, White, S. Asian, and Multi-racial.

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As noted previously, there was a wide range from university to university in office hours attendance, displayed in Table 14, with extremes of 34.0% of students at Univ. 3 and 76.6% of students at Univ. 2 not going to office hours; as with the S-OQs, we analyze the reasons for not attending office hours while controlling for university. In answer to S-RQ1, students chose "I did not need help" most frequently as a reason for not attending office hours at all universities, both in aggregate (65.7% of all students who did not attend office hours) and individually at each university. Table 16 shows which reasons students chose, the averages at each individual university, and the overall average. Tables 17, 18, and 19 show the percent of students choosing each reason by university and trait for S-NOH1, S-NOH3, and S-NOH4.[28]

	Reason	Type	Univ. 1	Univ. 2	Univ. 3	Univ. 4	Univ. 5	All
S-NOH1	I did not need any help.	internal	64.0	61.8	66.9	72.3	68.6	65.7
S-NOH2	The office hour times were inconvenient for my schedule.	external	37.0	34.8	34.6	23.5	31.1	34.1
S-NOH3	I did not feel comfortable with any of the TAs for this course.	external	8.0	6.2	5.2	7.8	2.8	6.1
S-NOH4	I did not believe the TAs would be able to help me with my needs for this course.	external	9.4	6.2	5.2	15.1	7.4	8.1

Table 16. Reasons students did not attend office hours (**S**tudent-**N**o **O**ffice **H**ours) with average selection rate per university

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Next, we explore the extent to which the reasons that students avoid office hours are due to internal factors versus external ones. We argue that a student thinking they do not need help is due to the internal factor of their own self-evaluation but leave deeper exploration of this topic for future work.[29] On the external side, the hours themselves being inconveniently scheduled is the second most popular choice, at 34.1% overall. While it might appear that this reason is neutral, neither an internally-avoidant nor externally-avoidant behavior, we argue that because these students might have wanted to go to office hours but couldn't due to external factors (e.g., a student who holds a job outside of the university may not be able to attend evening office hours), this is indicative of systemic/socioeconomic barriers and should be treated as an external factor. Our survey responses also show a meaningful portion of students avoid office hours even though they may be able to attend them–both in terms of not believing the TAs could help and in terms of not being comfortable with the TAs.

8.2.1 S-NOH1: "I did not need help." Table 17 shows highlighted results of the S-NOH1 analysis. When S-NOH1 was flagged as significant, men and students choosing "None" for disabilities[30] are more likely to pick this reason than their peers. When considering race alone, patterns are not identifiable between Univ. 1 and Univ. 2, the two universities flagged. However, within race and gender, we observe a pattern of men selecting this reason at a higher rate than their female peers whenever it is flagged (and in most cases when it is not).

Univ. 5 was the only institution where course was flagged as statistically significant for differences in proportions of student selecting S-NOH1. For this institution, students from CS 1 choose this reason (74.9%) more than CS 2 (65.2%) who choose it more than students from CS 3 (55.1%). This is the pattern we expect–as students progress in a program, courses become more difficult, making it more likely that students do need help.

[28]Full tables appear in Appendix C

[29]However, it is an important note that prior work has found that students in STEM disciplines may avoid office hours because of internal and external perceptions of help-seeking as weakness, which may lead students to indicate that they did not need help [78].

[30]Note that at Univ. 2, none of the students with disabilities attended office hours so we couldn't test difference of experience in office hours themselves–for this institution, we do see significant differences among reasons for not attending office hours.

factor	value	Univ. 1 %	Univ. 1 N	Univ. 2 %	Univ. 2 N	Univ. 3 %	Univ. 3 N	Univ. 4 %	Univ. 4 N	Univ. 5 %	Univ. 5 N
Gender	M	70.1*** (+)	668	68.6*** (+)	245	69.2	295	73.7	114	69.7	442
	Oth.	59.1*** (+)	22	38.5*** (+)	13					78.6	14
	W	52.7*** (+)	298	50.8*** (+)	124	61.3	80	76.5	34	64.9	131
Race	Black	54.8*** (+)	241	36.4*** (+)	11			60.0	15	59.1	44
	Arab/ME	62.5*** (+)	16			90.9	11			54.5	22
	White	74.2*** (+)	128	57.4*** (+)	47	70.0	217	79.0	81	70.4	216
	E. Asian	56.5*** (+)	124	62.0*** (+)	71	66.7	24			66.7	21
	Oth. Asian									52.6	19
	Other	63.2*** (+)	19	20.0*** (+)	10	54.5	11			66.7	24
	S. Asian	73.8*** (+)	271	75.0*** (+)	112	40.0	15	62.5	24	75.4	126
	SE. Asian	61.5*** (+)	109	65.0*** (+)	80	85.0	20			67.2	61
	Multi	66.7*** (+)	72	54.8*** (+)	42	66.0	53	61.5	13	76.6	47
Black intersectional	Black M	58.8	153					70.0	10	60.0	30
	Black W	46.8	79							50.0	12
White intersectional	White M	78.3	83	69.2	26	72.8	169	80.0	60	71.3	164
	White W	70.0	30	37.5	16	57.5	40	88.2	17	61.1	36
E. Asian intersectional	E. Asian M	66.7*** (+++)	87	68.2	44	68.4	19			66.7	12
	E. Asian W	33.3*** (+++)	30	56.5	23						
S. Asian intersectional	S. Asian M	78.9* (+++)	175	81.8* (+++)	66	40.0	10	65.0	20	76.8	99
	S. Asian W	64.6* (+++)	82	61.0* (+++)	41					76.0	25
SE. Asian intersectional	SE. Asian M	62.5	72	64.8	54	83.3	12			68.1	47
	SE. Asian W	54.5	33	65.0	20					64.3	14
Multi intersectional	Multi M	72.3	47	79.2*** (+++)	24	68.6	35			78.6	28
	Multi W	59.1	22	16.7*** (+++)	12	69.2	13			75.0	16
Eth.	Other Hisp.	65.2	46	36.4** (+)	11	55.6	18	70.0	10	65.6** (+)	64
	Cuban									83.3** (+)	12
	Chicanx	51.9	52	38.7** (+)	31	61.3	62	85.0	20	61.1** (+)	18
	Puerto Rican	58.3	12							37.5** (+)	16
	Not H/L/S	65.5	843	64.3** (+)	297	68.3	284	71.7	113	71.7** (+)	442
Dis.	None	65.7* (+++)	886	64.9*** (+++)	342	69.5	302	74.3	109	70.2	496
	Yes	54.7* (+++)	128	42.6*** (+++)	54	60.9	87	69.0	42	66.7	93

Table 17. Selected factors for S-NOH1, "I did not need help"; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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When we explore the significant difference in term observed at Univ. 5, we note that in Fall 2024, when S-NOH1 was chosen by the lowest percentage of students, response rates for students in CS 2 and CS 3 were much higher–86.7% versus 7.2% (FA2023) and 50% (SP2024) for CS 2 and 70.2% versus 6.5% (FA2023) and 0.8% (SP2024) for CS 3.

8.2.2 S-NOH2: "The office hour times were inconvenient for my schedule." On average, 34.1% of students selected S-NOH2. Course and term are the factors with the strongest results for students choosing S-NOH2, which corresponds with office hours times being set each term individually course by course. Observed effects are small and confidence values fall below 0.1 but not 0.01.

Race appears as significant at one university, providing a weak signal with no identifiable patterns. Race and gender are flagged for E. Asian students at Univ. 1 and White students at Univ. 4. For both of these groups, large effects sizes are observed, associated with women chosing this reason more than their male counterparts.

8.2.3 S-NOH3: "I did not feel comfortable with any of the TAs for this course." This option was selected at the lowest rate by students (6.1% average), which is, at first glance, heartening based on the fact that it indicates a serious problem. However, on closer inspection, students' likelihood of choosing S-NOH3 varies primarily based on ethnicity and

factor	value	Univ. 1 %	Univ. 1 N	Univ. 2 %	Univ. 2 N	Univ. 3 %	Univ. 3 N	Univ. 4 %	Univ. 4 N	Univ. 5 %	Univ. 5 N
Dis.	None	7.0	886	4.1** (+++)	342	4.6	302	3.7** (+++)	109	2.4	496
	Yes	11.7	128	13.0** (+++)	54	6.9	87	16.7** (+++)	42	5.4	93

Table 18. Percent students selecting S-NOH3 by disability status, "I did not feel comfortable with any of the TAs for this course" for disability status; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses for significant differences, - indicates insignificant, + small, ++ medium, and +++ large

disability status.[31] When inspecting these two factors individually, a clear pattern of historically marginalized students being more likely to indicate that they did not feel comfortable with any of the TAs emerges. This is a pattern that we see repeated when we consider S-NOH4–belief that TAs are able to help with students' needs.

Ethnicity is a significant factor at Univs. 1, 2, and 5. At these universities, students who indicate that they are not of Hispanic, Latinx, or Spanish origin choose S-NOH3 at a lower rate (6.8%, 4.7%, and 2.0% at Univs. 1, 2, and 5) than all of their peers with the singular exception of students at Univ. 5 where no Chicanx students chose this reason.

For disability status (Table 18), students who enter a disability are more likely to choose S-NOH3 than their peers at all universities; this is statistically significant with a large effect size at Univs. 2 and 4.

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factor	value	Univ. 1 %	Univ. 1 N	Univ. 2 %	Univ. 2 N	Univ. 3 %	Univ. 3 N	Univ. 4 %	Univ. 4 N	Univ. 5 %	Univ. 5 N
Gender	M	7.6** (-)	668	2.4*** (+)	245	4.1	295	12.3	114	7.9	442
	Oth.	0.0** (-)	22	15.4*** (+)	13					7.1	14
	W	12.8** (-)	298	11.3*** (+)	124	8.8	80	20.6	34	5.3	131
Race	Black	12.9** (+)	241	45.5*** (++)	11			33.3	15	6.8	44
	Arab/ME	18.8** (+)	16			0.0	11			13.6	22
	White	7.0** (+)	128	4.3*** (++)	47	5.1	217	13.6	81	8.8	216
	E. Asian	4.0** (+)	124	1.4*** (++)	71	12.5	24			4.8	21
	Oth. Asian									10.5	19
	Other	5.3** (+)	19	30.0*** (++)	10	0.0	11			4.2	24
	S. Asian	9.2** (+)	271	5.4*** (++)	112	0.0	15	16.7	24	10.3	126
	SE. Asian	5.5** (+)	109	1.2*** (++)	80	5.0	20			0.0	61
	Multi	15.3** (+)	72	9.5*** (++)	42	1.9	53	7.7	13	0.0	47
Multi intersectional	Multi M	12.8	47	0.0** (+++)	24	2.9	35			0.0	28
	Multi W	22.7	22	25.0** (+++)	12	0.0	13			0.0	16
Eth.	Other Hisp.	15.2** (+)	46	18.2	11	0.0	18	10.0	10	10.9** (+)	64
	Cuban									33.3** (+)	12
	Chicanx	3.8** (+)	52	0.0	31	6.5	62	10.0	20	11.1** (+)	18
	Puerto Rican	33.3** (+)	12							0.0** (+)	16
	Not H/L/SS	8.9** (+)	843	6.4	297	4.9	284	16.8	113	6.8** (+)	442
Dis.	None	9.3	886	5.3	342	3.3* (+++)	302	12.8	109	7.1	496
	Yes	7.8	128	11.1	54	9.2* (+++)	87	23.8	42	8.6	93

Table 19. Selected factors for S-NOH4, "I did not believe the TAs would be able to help me with my needs for this course"status; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

[31] Gender, race, and course all have insignificant to small effect sizes at individual universities. Intersectional race and gender are flagged for two groups with small to large effect sizes.

8.2.4 S-NOH4: "I did not believe the TAs would be able to help me with my needs for this course." Finally, our data shows that students with historically marginalized identities are more likely to choose this reason than their peers. When students do not go to office hours because they do not believe that the TAs are able to help with their needs. This indicates a serious and difficult problem to address. We highlight several significant factors in Table 19.

Women are more likely to choose this reason than their male counterparts at both universities where gender was identified as an important factor but note that the effect sizes are small to insignificant. When considering race, at Univ. 1, Arab/ME, Multi-racial, and Black students are most likely to pick this reason. At Univ. 2, the two groups with the lowest number of responses–Black students and those who entered something else for their racial identity–are most likely to choose this reason at 45.5% and 30.0%, respectively–though we note that these are also the two racial group with the lowest representation for Univ. 2. Univ. 1 provides an interesting observation–Black students select this reason significantly more than all other racial groups with large numbers of responses, even when they are not in a minority. Inspecting race and gender together, multi-racial students at Univ. 2 are the only group with a detectable difference–multi-racial men did not select this reason while 25% of multi-racial women did, though the numbers of respondents within these categories are relatively low, indicating an area for further consideration in the future.

Next, we turn to ethnicity and disability status. The trends within disability status and ethnicity are not as clean-cut as they were for S-NOH3. For disability status, flagged at Univ. 3 with a large effect size, students who indicate a disability are almost three times more likely to choose S-NOH4 than students who indicate no disability. Speaking to ethnicity, trends show that for both schools flagged, students who are not of Hispanic, Latinx, or Spanish origin are in the two groups least likely to pick S-NOH4. Zooming out, there remains a trend that students with historically marginalized identities are more likely to choose S-NOH4.

While course was flagged at Univ. 1 and Univ. 5, there are no identifiable patterns.

As shown in Table 16, there is large variance university to university with regards to non-attendance due to mistrust in TAs. Not only does this make controlling for university particularly prescient, but gives a strong signal to Univ. 4–where 15.1% of students not attending office hours selected S-NOH4–that one barrier their institution is facing for improving culture in their computing classrooms is mistrust in their TAs. Notably, no demographic factors were flagged at Univ. 4 for S-NOH4, indicating that the mistrust is shared among students of all identities.

8.2.5 Summary. As a whole, our investigation shows that students with historically marginalized identities in computing are more likely to avoid office hours due to external factors versus their peers. Even at our school with the highest proportion of responses from Black students, they show a higher degree of mistrust in their TAs than their white and Asian peers. We further find that when race and gender are considered intersectionally, we see an indication that women might be externally-avoidant at higher rates than their male peers, but sparsity prevents us from drawing a firm conclusion. We further observe that that Hispanic, Latinx, and students of Spanish origin are more likely to avoid office hours due to these external factors. Finally, the picture for students indicating they have a disability or choosing to skip the question is bleak–not only are they more likely to have a worse time in office hours than their peers, shown in Table 15, they are also more likely to avoid office hours because they aren't comfortable with the TAs.

9 Discussion and Conclusions

In this section, we first summarize the contributions of our study, then discuss our findings in detail before finally giving three overarching recommendations.

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With this study, we create a landscape portrait of the experiences of students and TAs in computing departments at a diversity of institutions across the U.S. Our quantitative approach contributes a unique view of the TA ecosystem as a whole. By collecting data over two years at five different institutions, we gathered a dataset that is larger than any previous dataset evaluating TA experiences, student-TA dynamics, or TA training in computing programs, enabling us to reveal insights previously unavailable. With this work, we show that TAs benefit from receiving TA training, that student disability status and race have the largest impacts on their experiences in office hours and that students with historically marginalized identities are more likely to avoid office hours due to factors outside of their control, causing them to be more *externally*-avoidant. These results are particularly prescient for computing programs who rely on TAs to provide support to students, especially in introductory courses as well those who are engaged in BPC projects. They also stand in support of devoting resources to TA training programs and careful thought to the barriers that students may experience when accessing office hours.

TA training: The overall goal of this study from the TA perspective was to determine whether or not in-person, pedagogically-focused TA training had significant impacts on TA experience. Our data did not show this was the case; we are therefore unable to reject the null hypothesis of TA-RQ3, that in-person, pedagogy-focused training impacts TAs differently than logistics-only, asynchronous, or no training. However, our results *do* show that if an institution trains their TAs at all, the TAs are more comfortable and confident in their skills and duties, supporting previous findings that training generally makes TAs feel more prepared [20, 50, 75].

Based on what has been observed in previous studies, we believe that specifically in-person training is likely better than asynchronous, online training [7, 33, 50, 75]. Our hypothesis is that our intervention wasn't large enough to have effects that were detectable or that the impacts had normalized by the end of the term when surveys were conducted. These trainings, which prioritized feasibility given the financial and structural constraints that universities work under, ranged from as short as two hours to as long as eight hours. [32]

TA experience: In response to our other research questions, this work paints a vivid picture of what factors influence TA experience. For TA-RQ1, we discovered that the most important factors for TA comfort and confidence are race, course, university, class standing, and whether or not TAs had any training. White, S. Asian, undergraduate, and CS 1 TAs tend to report higher perceptions of their own skills and comfort with office hours. We also uncovered quantitative evidence that E. Asian, S. Asian, and SE. Asian TAs have different experiences. While this is not surprising based on previous literature [15, 45, 49, 71, 72], our study stands as continued support for disaggregating racial categories whenever possible.[33] Perhaps counterintuitively, on average, Ph.D. students report the lowest confidence in their subject knowledge and ability to explain concepts.

In terms of perception of TA training (TA-RQ2), race, course, and class standing were the most significant factors. In spite of the differences time devoted to employing the TA training from the experiment, we found that university is not one of the factors that influences TAs' reported perceptions of TA training. When we inspected class standing, we saw that MS students have the highest ratings of TA training. Further inspection of the demographics of MS students shows that S. Asian students are a majority of this population, indicating this may be the same effect that we observed when we studied race in isolation–S. Asian TAs consistently rated all questions higher than their peers. This is consistent with the results of our question about how much training TAs prefer–there is evidence that race is the most influential

[32]We made this choice so that a greater diversity of schools would be eligible to participate because of the paucity of studies on TAs and student-TA interactions in diverse environments.

[33]Racial identity categories collected by certain governmental agencies such as IPEDS aggregate all Asian students into a single category: https://nces.ed.gov/ipeds/report-your-data/race-ethnicity-definitions.

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factor, given that class standing was *not* significant for this question. Once again, we note that because of demographic sparsity, we are unable to rule intersectional race and gender as an influencing factor out entirely. One more potential confounding factor to this result is that there may be cultural bias toward S. Asian students generally giving higher ratings in surveys.

While prior work evaluating TA experiences from their own perspectives was done through a qualitative lens–identifying elements like classroom layout [65] and specific stressors that TAs experience, such as long queues and pressure to give students solutions [52]–our quantitative analysis adds to this previous work, further expanding our understanding of TA experiences from a scaled-up perspective and bringing to light the correlations between TA identity and experience.

Students: In answer to S-RQ1, whether or not students are attending office hours, and, if they aren't attending, why not, we find that students who are much likelier to not attend office hours because they think they do not need help tend to belong to two groups: 1) men over their non-male peers, a finding that holds when doing intersectional race and gender analysis, and 2) students who indicate they do not have a disability over students who indicate that they do. When considering external reasons for not attending office hours, such as not believing the TAs would help, we find that, as a whole, students with historically marginalized identities are more likely to choose these options. In this work, we defined externally-avoidant as being pushed away by external factors while internally-avoidant is avoiding office hours because of internal factors, such as feeling confident in the material.

Of particular note is our analysis on students with disabilities. These students are externally-avoidant more often than their peers. We also find strong evidence that when students with a disability *do* attend office hours, they have worse experiences than their peers. This confirms that the fears students with disabilities have around going to office hours are unfortunately well-founded.

Rounding out our findings for S-RQ2, what factors contribute to student experience in office hours with TAs, other influential factors include race, course, and term. Within these factors, we corroborate the statement that where possible, race should not be evaluated in aggregated categories–E. Asian, S. Asian, and SE. Asian students report significantly different experiences in office hours, with E. Asian students giving lower scores in the majority of cases. Along the dimension of course, CS 1 students rate their experiences lower than their CS 2 counterparts in all cases.

Because prior work on student evaluations of TAs has worked with student evaluations of specific TAs, rather than student experiences with TAs in general, the conclusions we can draw are once again complementary. While prior work identifies that that TA gender does not correlate with differences in student evaluation [67, 80], this experiment shows how *student* demographic attributes impact their experiences and choices about whether or not to go to office hours.

Recommendations: We make three recommendations for computing departments based on the evidence shown in our experiment. These recommendations are particularly prescient for departments who rely on TAs, as the evidence in this study shows that students with historically marginalized identities are more likely to avoid office hours due to external factors.

(1) We find that across five universities and contexts, training TAs increases TA comfort and confidence. While our results were not able to differentiate between logistics-only training and our intervention training, prior work indicates that interpersonal training is necessary for supporting student sense of belonging. We therefore recommend providing TA training that includes modules addressing such topics, including the following.
   - Mentorship and connecting to students [57, 61, 63, 70].

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- Effectively asking questions and guiding students without giving answers [55, 63, 70].
- Pedagogy and mindset [2, 70].
- Listening skills [5].

(2) Disability *must* be considered when assessing student experience in office hours based on our findings which show that students with disabilities are more reticent to attend office hours and report having worse experiences when they do. We recommend that institutions specifically consider accessibility needs when structuring office hours, while we hope future insight into disability and office hours in computing specifically can provide more specific recommendations, we note some recommendations from accessibility studies below.
- Emphasizing listening skills in TA training [5].
- Ensuring office hours are in easily physically accessible locations [5, 13].
- Considering sensory aspects of office hours locations, such as sound and light levels, choosing quieter locations when possible [25].

(3) Race *must* be considered when evaluating student-TA interactions and designing interventions. We found in our analysis of TA comfort and perceptions of training as well as our analysis of student experience with office hours that race was an important factor in impacting TA and student experience. We conclude that that cultural context must be considered when developing TA trainings. In addition to interviewing students at your specific institution to better understand your context, one way to approach this is to develop mentorship programs–so TAs and students might develop stronger rapport and senses of belonging [53].

While we did not see consistent differences in student experience based on gender alone, we do see difference in comfort with and trust in the TAs for those who do *not* attend office hours along gendered lines. Designing an intervention to improve this situation must explicitly work to improve the office hours experience for these students–rather than simply attempting to increase the rate at which they attend office hours.

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10 Future Work

In the future, it would be useful to re-run this experiment with a few important tweaks. To speak to experimental design, we prioritized ease of launch at many universities and therefore had few requirements for TA training programs other than they be in-person, pedagogically-focused trainings that drew from the same collection of modules. In the future, we would design a study that has more requirements around specific length and therefore also more specific content in the training provided. Additionally, we would gather the training teams at each university together for a larger, coordinated "train the trainers" session to facilitate uniformity of the TA training programs launched. One of the difficulties that led to fewer requirements on specific TA training program content, other than amount of time devoted to training, was the specific political climates that each university was operating under. Finally, in the most ideal world, we would gather an entire year of baseline data from participating universities so that temporal comparisons could be made between corresponding semesters–falls with falls and springs with springs.

Next, we hypothesize that one reason we do not see differences for TAs along the lines of experience, amount of training, and kind of training is because by the time the TAs took the survey at the end of the term, new TAs' experience levels had grown enough to normalize with the rest of the TA population. In the future, it would be useful to take data points earlier the term–specifically by having a survey closer to the time when TAs were trained. Such a study would also allow for longitudinal evaluation of how TA confidence in their own abilities shifts as the term progresses.

Finally, there are many other questions that we could not sufficiently explore in this study–how do faculty experiences with TAs correspond to student and TA experiences? What can evaluation of free response data tell us that is not indicated in quantitative evaluation of survey responses? What does qualitative data from TAs across institutions tell us about what TAs find most useful from TA training? How do the trends we observe in this study correlate with the similarities and differences between TA and student demographics?

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A Participant information sheets

A.1 TA Participant Sheet

University and Department: Northeastern University, Computer Science
Name of Investigator(s): Carla Brodley, Felix Muzny
Title of Project: Multi-University Teaching Assistant Training in Computing

Request to Participate in Research

We would like to invite you to take part in a research project. The purpose of this research is to evaluate the effects of TA training programs in computing departments and courses.

You must be at least 18 years old to be in this research project.

The study will take place online and will take about 5 – 30 minutes. If you decide to take part in this study, we will ask you to fill out two surveys[34], one at the beginning of term and one at the end of term to discuss your opinions about your experiences as a TA at your institution.

The possible risks or discomforts of the study are minimal. You may feel a little uncomfortable answering personal questions.

There are no direct benefits to you for participating in the study. However, your answers may help us to learn more about developing and improving the experiences of students, TAs, and faculty in computing departments.

Your part in this study is anonymous. That means no one will know if you took part in this study and no one, including the researcher, will know what your answers are. Any reports or publications based on this research will use only group data and will not identify you or any individual as being of this project.

It is possible that respondents could be identified by the IP address or other electronic record associated with the response. Neither the researcher nor anyone involved with this survey will be capturing those data. If you have any questions regarding electronic privacy, please feel free to contact UNIVERSITY's Office of Information Security via phone at ANONYMIZED, or via email at ANONYMIZED.

The decision to participate in this research project is up to you. You do not have to participate and you can refuse to answer any question. Even if you begin the study, you may withdraw at any time.

You will not be directly paid for your participation in this study. If you are compensated hourly, you may be compensated by your home institution by counting the time you spend filling these surveys out as time worked.

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[34]Though we collected data at both the beginning and the end of term, we only analyze data collected at the end of term in this work.

If you have any questions about this study, please feel free to email ANONYMIZED. If you have any questions about your rights in this research, you may contact the Human Subject Research Protection, ANONYMIZED. You may call anonymously if you wish. This study has been reviewed and approved by the ANONYMIZED Institutional Review Board (# xx-xx-xx).

You may keep this form for yourself.

Thank you.

A.2 Student Participant Sheet

University and Department: Northeastern University, Computer Science
Name of Investigator(s): Carla Brodley, Felix Muzny
Title of Project: Multi-University Teaching Assistant Training in Computing

Request to Participate in Research

We would like to invite you to take part in a research project. The purpose of this research is to evaluate the effects of TA training programs in computing departments and courses.

You must be at least 18 years old to be in this research project.

The study will take place online and will take about 5 – 15 minutes. If you decide to take part in this study, we will ask you to fill out two surveys, one at the beginning of term and one at the end of term to discuss your opinions about your experiences as a TA at your institution.

The possible risks or discomforts of the study are minimal. You may feel a little uncomfortable answering personal questions.

There are no direct benefits to you for participating in the study. However, your answers may help us to learn more about developing and improving the experiences of students, TAs, and faculty in computing departments.

Your part in this study is anonymous. That means no one will know if you took part in this study and no one, including the researcher, will know what your answers are. Any reports or publications based on this research will use only group data and will not identify you or any individual as being of this project.

It is possible that respondents could be identified by the IP address or other electronic record associated with the response. Neither the researcher nor anyone involved with this survey will be capturing those data. If you have any questions regarding electronic privacy, please feel free to contact UNIVERSITY's Office of Information Security via phone at ANONYMIZED, or via email at ANONYMIZED.

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The decision to participate in this research project is up to you. You do not have to participate and you can refuse to answer any question. Even if you begin the study, you may withdraw at any time.

You will not be paid for your participation in this study.

If you have any questions about this study, please feel free to email ANONYMIZED. If you have any questions about your rights in this research, you may contact the Human Subject Research Protection, ANONYMIZED. You may call anonymously if you wish. This study has been reviewed and approved by the ANONYMIZED Institutional Review Board (# xx-xx-xx).

You may keep this form for yourself.

Thank you.

B Surveys

B.1 TA Survey

Questions with direct correlates to the student survey from CERP are indicated with +s at the end of the question.

(1) What is your class standing?
- Undergraduate student
- Master's student
- Ph.D. student

(2) If the answer to the previous question was undergraduate student: What is your programming experience prior to this term?
- None
- I have taken one programming course
- I have taken 2 - 4 programming courses
- I have taken five or more programming courses
- I have a small amount of self-taught programming experience
- I have a medium amount of self-taught programming experience
- I have a large amount of self-taught programming experience

(3) Course TAing What is the course code for the course that you are TAing this term?

(4) Have you been a TA previously?
- Yes, at a different institution (not your current institution)
- Yes, at my current institution
- No

(5) If the answer to the previous question was "Yes": How many terms (not including this one) were you a TA for previously?

(6) Rate the amount of programming experience that you had before attending university as an undergraduate.
- None

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- A small amount - I had done or seen some programming but hadn't worked with it more than a few hours
- Some - I had done some programming but less than following an entire course
- A significant amount - I took a programming course (or equivalent)

Please think about the students in the course that you are currently TAing, and indicate how true each statement is based on your experience, observations and assumptions. All questions rated on the scale:

- Not at all true
- Slightly true
- Moderately true
- Quite true
- Extremely true

(1) The students in the course I'm TAing have programmed before.
(2) The students in the course I'm TAing are computing majors.
(3) The students in the course I'm TAing generally look like me.
(4) The students in the course I'm TAing have a similar academic background to me.
(5) The students in the course I'm TAing have a similar cultural background to me.
(6) I am knowledgeable about the course content for which I am a TA.+
(7) I can explain course concepts clearly for the course content for which I am a TA.+
(8) I know how to encourage questions and engagement.+
(9) I know how to create a welcoming environment in office hours.+
(10) I am comfortable telling someone that I don't know the answer to a question.
(11) I am comfortable conducting office hours.

TA experiences:

(1) In your opinion, which of the following has prepared you the best for your job as a TA?
- General knowledge of the subject material for the course I'm TAing
- Experience completing the assignments for the course I'm TAing
- Earning a high grade in the course I'm TAing or an equivalent course
- Prior experience teaching or tutoring
- Prior experience from working a different job
- Training the professor teaching the course I'm TAing has provided
- Training that has been provided by my department or university
- I do not feel prepared

(2) As a student, which statement below best describes how often you generally interact with TAs in the computing courses that you've taken previously?'
- I never interact with TAs in the computing courses that I've taken
- I rarely interact with TAs in the computing courses that I've taken
- I sometimes interact with TAs in the computing courses that I've taken
- I frequently interact with TAs in the computing courses that I've taken
- Not applicable - previous institution had no TAs

(3) Which statement below best describes the impact TAs in computing courses have had on your interest in computing generally?+

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- My own experience with TAs has made me significantly less interested in computing
- My own experience with TAs has made me less interested in computing
- My own experience with TAs has had no impact on my interest in computing
- My own experience with TAs has made me more interested in computing
- My own experience with TAs has made me significantly more interested in computing

(4) Which statement below best describes the impact TAs in computing courses have had on your interest in teaching?+
- My own experience with TAs has made me significantly less interested in teaching
- My own experience with TAs has made me less interested in teaching
- My own experience with TAs has had no impact on my interest in teaching
- My own experience with TAs has made me more interested in teaching
- My own experience with TAs has made me significantly more interested in teaching

(5) Which statement below best describes the impact professors in computing courses have had on your interest in teaching?+
- My own experience with professors has made me significantly less interested in teaching
- My own experience with professors has made me less interested in teaching
- My own experience with professors has had no impact on my interest in teaching
- My own experience with professors has made me more interested in teaching
- My own experience with professors has made me significantly more interested in teaching

(6) Based on your experience working as a TA so far, rate how likely you are to pursue TAing in the future.
- It is very unlikely that I will pursue being a TA in the future
- It is unlikely that I will pursue being a TA in the future
- It is neither unlikely nor likely that I will pursue being a TA in the future
- It is likely that I will pursue being a TA in the future
- It is very likely that I will pursue being a TA in the future
- Not applicable - I am graduating

(7) What best describes the most difficult situation you have encountered as a TA so far this term?
- Managing grading or dealing with student push back on grading
- Explaining information without giving away the answers to questions
- Answering questions when I don't fully understand a concept myself
- Helping students who have foundational knowledge gaps
- Responding to student behavior (rudeness, anxiety, etc.) and setting boundaries
- Practicing good time management, especially in office hours

(8) Select all kinds of TA training you received this term. Do not include weekly course meetings.
- In person or synchronous, provided by my department
- In person or synchronous for just the course I'm TAing
- Asynchronous online training
- I did not receive TA training this term
- Something else: TEXT BOX HERE

(9) Select all topics covered in the training you received.
- University policies

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- Employment logistics
- Teaching strategies
- Grading
- Course-specific logistics and policies
- Interacting with students
- Approaching learners with different backgrounds
- Something else: TEXT BOX HERE

Please think about your experiences attending TA training this term and indicate how true each statement below is. All questions rated on the scale:

- Not at all true
- Slightly true
- Moderately true
- Quite true
- Extremely true

(1) I found TA training helpful.
(2) TA training made me more confident performing my duties as a TA.
(3) TA training made me more comfortable performing my duties as a TA.
(4) TA training changed how I approach students in the course I am TAing.
(5) TA training changed how I communicate with the professor I am working with.

Training amount and open-ended questions:

(1) Which statement best describes how you feel about the amount of training you received?
    - I would prefer significantly more training as a TA
    - I would prefer more training as a TA
    - I am happy with the current amount of training
    - I would prefer less training as a TA
    - I would prefer significantly less training as a TA
(2) Were there particular situations or duties in your job as a TA that you felt well prepared for? What were they?
(3) Were there any situations or duties in your job as a TA that you wish you had been better prepared for? what were they?
(4) Do you have any other feedback that you'd like to share with the TA training team?

B.2 Student Survey

Questions that came directly from CERP are indicated with *s at the end of each question.

(1) What is the course code and number that you are filling this survey out for?
(2) How many times did you attend TA office hours for this course during this term?*
    - Not at all
    - Once
    - Two or three times
    - Four or five times

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- More than five times

(3) If the answer to the previous question was "Not at all": Why didn't you attend office hours for this course during this term? Please check all that apply.*
- I did not need any help.
- The office hour times were inconvenient for my schedule.
- I did not feel comfortable with any of the TAs for this course.
- I did not believe the TAs would be able to help me with my needs for this course.
- Other; please describe your reason(s):

(End of survey)

(4) How many different TAs did you see in office hours during this term?*
- One
- Two or three
- Four or five
- More than five

(5) If the answer to the previous question was more than one: In general, how much did your experiences vary depending on which TA you saw?*
- My experiences with different TAs were similar
- My experiences with different TAs were somewhat different
- My experiences with different TAs were very different

For the questions that follow, please think about the experience(s) you had when you visited TA(s) during their office hours for this course, and indicate how true each statement is based on your experience.
If you had different experiences on different occasions, please report an average of all of your experiences with TAs during office hours for this course. All questions rated on the scale:

- Not at all true
- Slightly true
- Moderately true
- Quite true
- Extremely true

(1) The TAs were knowledgeable about the course content.*
(2) The TAs explained course concepts clearly.*
(3) The TAs encouraged my questions and engagement.*
(4) The TAs gave clear explanations to my questions.*
(5) The TAs were respectful to me.*
(6) The TAs created a welcoming environment.*
(7) I felt comfortable asking for help from the TAs.*
(8) The TA office hours were easy to access.*
(9) I felt that the TAs wanted me to succeed.*
(10) I would feel comfortable asking the TAs for advice about academic matters not related to this course.*

Overall:

(1) How would you describe your overall experience(s) with TAs during office hours for this course?*

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- Very negative
- Slightly negative
- Neither negative nor positive
- Slightly positive
- Very positive

(2) Overall, how successful were TAs in creating an environment that was conducive to learning during office hours for this course?*
- Very negative
- Slightly negative
- Neither negative nor positive
- Slightly positive
- Very positive

(3) How would you rate the overall effectiveness of the TAs' instruction and support during office hours for this course?*
- Not at all effective
- Slightly effective
- Moderately effective
- Quite effective
- Extremely effective

(4) Which statement below best describes the TAs' impact on you?*
- My experience with the TAs made me less interested in computing.
- My experience with the TAs had no impact on my interest in computing.
- My experience with the TAs made me more interested in computing.

(5) If there was any particularly positive or negative experience that you had with a TA during office hours for this course that you would like to share, please do so below.*

(6) Please provide any additional comments below that you would like to share about your experiences with your TAs for this course.*

These questions are then followed by demographics questions, included in Appendix B.3.

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B.3 Demographic Questions

Demographic questions were the same in all surveys deployed. They were asked at the end of the survey.

Introduction: Thank you for answering these questions. We are almost done. This last section asks a few questions about you and should go by quickly. Why should you provide this information? People experience things in different ways, and these details help us understand who and why. Demographics help us understand the experiences and perspectives of people with different identities. If you prefer not to answer a given question, simply skip that question.

(1) In what year did you enter this institution for your current program?

(2) What is your major, or one of your majors?

(3) Do you have a second major? If yes, what is it?

(4) What type of disability do you have? Please check all that apply.

- I do not have a disability
- Attention deficit
- Auditory disability
- Autism spectrum disorder
- Blind or visually impaired (does not include near/far sightedness)
- Health-related disability
- Learning disability
- Mental health disability
- Mobility or orthopedic disability
- Speech or language disorder
- Something else: TEXT BOX HERE

(5) What is your gender identity? Select all that apply.[35]
- Agender
- Cisgender
- Genderqueer/gender non-conforming
- Man
- Non-binary
- Transgender
- Woman
- Something else: TEXT BOX HERE

(6) What is the highest level of education attained by any of your parent(s)/guardian(s)?
  (a) Less than high school
  (b) High school graduate or G.E.D.
  (c) Some college (no degree earned)
  (d) Associate's degree
  (e) Bachelor's degree
  (f) Master's degree
  (g) Doctoral degree
  (h) I don't know

(7) What is your ethnicity? Select all that apply.
- Cuban
- Mexican, Mexican American, Chicanx
- Puerto Rican
- Another Hispanic, Latinx, or Spanish origin
- Not of hispanic, Latinx, or Spanish origin

(8) What is your race? Select all that apply.
- African American/African/Black
- Arab/Middle Eastern

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[35]The gender identity question was phrased this way so that transgender and gender-variant participants would feel less pressure to answer a question that explicitly states "Do you identify as transgender?"–in particular because local regulations and attitudes in the United States might make this a dangerous piece of information to divulge. This decision was made partly based on the author's lived experience. We note that the vast majority of participants chose "Man" or "Woman" and did not also choose "Cisgender" or another additional identity term.

- Caucasian/European/White
- East Asian (e.g., Chinese, Japanese, Korean)
- Indigenous or First Nations
- Native American/Alaska Native
- Native Hawaiian/Pacific Islander
- Southeast Asian (e.g., Cambodian, Vietnamese, Hmong, Filipino)
- South Asian (e.g., Indian, Pakistani, Nepalese, Sri Lankan)
- Other Asian
- Something else: TEXT BOX HERE

C Full Results Tables

Included in this appendix are full results tables for the mean values of the survey questions studied. TA-CQ questions are found in Table 20, TA-TQ questions in Table 21, S-OQ1 in Table 22, S-OQ2 in Table 23, S-OQ3 in Table 24, student office hours attendance in Table 25, S-NOH1 in Table 26, S-NOH2 in Table 27, S-NOH3 in Table 28, and S-NOH4 in Table 29.

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C.1 Full TA Results

factor	value	N	TA-CQ 1	TA-CQ 2	TA-CQ 3	TA-CQ 4	TA-CQ 5	TA-CQ 6	Avg.
Gender	M	161	4.6	4.5	4.2	4.4	4.3	4.6	4.4
	W	88	4.4	4.4	4.1	4.4	4.3	4.4	4.3
Race	Black	12	4.2^{***} (+++)	4.1^{***} (++)	3.8^{**} (++)	4.6	4.6^{*} (+)	4.7^{*} (+)	4.3^{**} (++)
	Arab/ME	16	3.8^{***} (+++)	3.8^{***} (++)	3.8^{**} (++)	4.3	4.6^{*} (+)	4.5^{*} (+)	4.1^{**} (++)
	White	57	4.7^{***} (+++)	4.6^{***} (++)	4.2^{**} (++)	4.3	4.2^{*} (+)	4.5^{*} (+)	4.4^{**} (++)
	E. Asian	28	4.3^{***} (+++)	4.3^{***} (++)	4.0^{**} (++)	4.2	4.0^{*} (+)	4.4^{*} (+)	4.2^{**} (++)
	S. Asian	100	4.6^{***} (+++)	4.6^{***} (++)	4.4^{**} (++)	4.5	4.3^{*} (+)	4.7^{*} (+)	4.5^{**} (++)
	SE. Asian	13	4.7^{***} (+++)	4.4^{***} (++)	3.9^{**} (++)	3.9	4.8^{*} (+)	4.1^{*} (+)	4.3^{**} (++)
	Multi	18	4.1^{***} (+++)	4.1^{***} (++)	3.9^{**} (++)	4.2	4.3^{*} (+)	4.2^{*} (+)	4.1^{**} (++)
White intersectional	White M	31	4.7	4.5	4.3	4.4	4.1	4.5	4.4
	White W	20	4.7	4.7	4.2	4.3	4.3	4.5	4.5
E. Asian intersectional	E. Asian M	14	4.3	4.4	4.1	4.2	4.3	4.4	4.3
	E. Asian W	12	4.3	4.4	3.9	4.2	3.8	4.4	4.2
S. Asian intersectional	S. Asian M	66	4.6	4.6	4.4	4.5	4.3	4.7	4.5
	S. Asian W	32	4.6	4.6	4.4	4.5	4.4	4.6	4.5
Eth.	Other Hisp.	11	4.4	4.2	4.1	4.3	4.1	4.3	4.2
	Chicanx	13	4.6	4.5	4.2	4.6	4.2	4.8	4.5
	Not H/L/S	207	4.5	4.4	4.2	4.3	4.3	4.5	4.4
Dis.	None	230	4.5	4.4	4.2	4.4	4.3	4.5	4.4
	Yes	22	4.6	4.5	4.1	4.3	4.4	4.3	4.4
Course	CS0	17	4.5^{***} (++)	4.4^{***} (++)	4.2^{*} (+)	4.4^{*} (+)	4.4	4.5	4.4^{*} (+)
	CS1	104	4.7^{***} (++)	4.6^{***} (++)	4.3^{*} (+)	4.3^{*} (+)	4.3	4.5	4.4^{*} (+)
	CS2	59	4.5^{***} (++)	4.5^{***} (++)	4.1^{*} (+)	4.4^{*} (+)	4.4	4.4	4.4^{*} (+)
	CS3	24	4.5^{***} (++)	4.7^{***} (++)	4.6^{*} (+)	4.7^{*} (+)	4.4	4.8	4.6^{*} (+)
	Discrete 1	27	4.3^{***} (++)	4.2^{***} (++)	4.0^{*} (+)	4.1^{*} (+)	4.2	4.4	4.2^{*} (+)
	Other	82	4.1^{***} (++)	4.0^{***} (++)	4.0^{*} (+)	4.3^{*} (+)	4.2	4.5	4.2^{*} (+)
Univ.	1	50	4.5^{*} (+)	4.5^{**} (+)	4.4	4.5	4.3	4.6^{***} (++)	4.5
	2	60	4.6^{*} (+)	4.4^{**} (+)	4.2	4.1	4.3	4.0^{***} (++)	4.3
	3	88	4.6^{*} (+)	4.6^{**} (+)	4.1	4.4	4.3	4.5^{***} (++)	4.4
	4	72	4.4^{*} (+)	4.4^{**} (+)	4.2	4.4	4.3	4.7^{***} (++)	4.4
	5	43	4.1^{*} (+)	4.0^{**} (+)	4.0	4.3	4.3	4.5^{***} (++)	4.2
Term	SP2023	32	4.6	4.5	4.3	4.4	4.1	4.4	4.4
	FA2023	139	4.4	4.3	4.1	4.3	4.4	4.5	4.3
	SP2024	59	4.5	4.5	4.2	4.4	4.3	4.6	4.4
	FA2024	83	4.5	4.5	4.2	4.4	4.3	4.4	4.4
class standing	UG	160	4.6^{**} (+)	4.5^{*} (+)	4.1^{*} (+)	4.3	4.4	4.4	4.4
	MS	79	4.5^{**} (+)	4.4^{*} (+)	4.3^{*} (+)	4.4	4.2	4.6	4.4
	Ph.D.	72	4.2^{**} (+)	4.2^{*} (+)	4.0^{*} (+)	4.3	4.3	4.6	4.3
New TA	experienced	220	4.5	4.4	4.2	4.4	4.4	4.5	4.4
	new	93	4.4	4.3	4.1	4.2	4.2	4.4	4.3
Training	not in person	151	4.4	4.4	4.0	4.3	4.2	4.4	4.3
	in person	162	4.5	4.4	4.2	4.4	4.4	4.5	4.4
Training async	no async training	225	4.5	4.5	4.2	4.3	4.3	4.5	4.4
	received async training	88	4.4	4.3	4.2	4.4	4.3	4.5	4.4
Training in person + async	did not receive both	274	4.5	4.4	4.2	4.3	4.3	4.5	4.4
	received both	39	4.4	4.3	4.2	4.4	4.4	4.5	4.4
Training	trained at all	211	4.5	4.4	4.2^{*} (+++)	4.4^{*} (+++)	4.4^{*} (+++)	4.5^{*} (+++)	4.4^{*} (+++)
	no training	102	4.4	4.4	4.0^{*} (+++)	4.1^{*} (+++)	4.1^{*} (+++)	4.3^{*} (+++)	4.2^{*} (+++)
baseline/intervention	intervention	190	4.5	4.5	4.2	4.4	4.3	4.5	4.4
	baseline	123	4.5	4.3	4.1	4.3	4.2	4.5	4.3

Table 20. Mean values for TA-CQ questions for every tested dimension; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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factor	value	N	TA-TQ 1	TA-TQ 2	TA-TQ 3	TA-TQ 4	TA-TQ 5	TA-AmQ
Gender	M	161	3.9	3.8	3.8	3.6	3.4	3.1
	W	88	3.8	3.6	3.7	3.4	3.2	3.2
Race	Black	12	4.0*** (++)	4.0** (++)	3.9*** (++)	3.5** (++)	3.5** (++)	3.4** (++)
	Arab/ME	16	3.7*** (++)	3.6** (++)	3.8*** (++)	3.4** (++)	3.2** (++)	2.6** (++)
	White	57	3.5*** (++)	3.4** (++)	3.3*** (++)	3.1** (++)	2.9** (++)	3.1** (++)
	E. Asian	28	3.6*** (++)	3.6** (++)	3.5*** (++)	3.4** (++)	3.2** (++)	3.1** (++)
	S. Asian	100	4.2*** (++)	4.1** (++)	4.2*** (++)	3.9** (++)	3.7** (++)	3.2** (++)
	SE. Asian	13	3.2*** (++)	3.1** (++)	3.0*** (++)	3.1** (++)	3.4** (++)	3.2** (++)
	Multi	18	3.4*** (++)	3.4** (++)	3.3*** (++)	3.2** (++)	2.7** (++)	3.5** (++)
White intersectional	White M	31	3.7	3.5	3.4	3.2	3.0	3.0
	White W	20	3.3	2.9	3.0	2.9	2.5	3.0
E. Asian intersectional	E. Asian M	14	0.0	0.0	0.0	0.0	0.0	3.1
	E. Asian W	12	0.0	0.0	0.0	0.0	0.0	3.1
S. Asian intersectional	S. Asian M	66	4.2	4.0	4.1	3.9	3.7	3.1
	S. Asian W	32	4.4	4.3	4.3	3.9	3.8	3.3
Eth.	Other Hisp.	11	3.8	3.2	3.1	3.0	3.2	3.2
	Chicanx	13	4.0	3.7	3.9	3.4	2.4	3.5
	Not H/L/S	207	3.8	3.7	3.7	3.5	3.4	3.1
Dis.	None	230	3.9	3.8	3.7	3.5	3.4	3.1
	Yes	22	3.6	3.4	3.4	3.2	2.7	2.8
Course	CS0	17	4.1	4.0* (+)	3.9* (+)	3.7* (+)	3.0* (++)	3.2* (+)
	CS1	104	3.8	3.7* (+)	3.7* (+)	3.5* (+)	3.4* (++)	3.2* (+)
	CS2	59	4.1	4.1* (+)	4.2* (+)	3.9* (+)	3.8* (++)	3.3* (+)
	CS3	24	4.1	3.8* (+)	3.8* (+)	3.6* (+)	3.7* (++)	3.2* (+)
	Discrete 1	27	3.3	3.1* (+)	3.2* (+)	3.1* (+)	2.7* (++)	3.1* (+)
	Other	82	3.8	3.6* (+)	3.6* (+)	3.2* (+)	3.0* (++)	2.9* (+)
Univ.	1	50	4.1	3.9	4.0	3.8	3.8	3.3
	2	60	3.8	3.8	3.7	3.7	3.3	3.0
	3	88	3.6	3.5	3.5	3.3	3.0	3.2
	4	72	4.0	3.8	3.9	3.7	3.4	3.1
	5	43	3.8	3.7	3.6	3.2	3.2	3.0
Term	SP2023	32	4.0	3.9	3.9	3.8	3.6	2.8
	FA2023	139	3.8	3.6	3.7	3.4	3.1	3.1
	SP2024	59	3.9	3.8	3.8	3.7	3.5	3.2
	FA2024	83	3.9	3.9	3.8	3.6	3.7	3.2
class standing	UG	160	3.6** (+)	3.6* (+)	3.5** (+)	3.3** (+)	3.2** (+)	3.2
	MS	79	4.2** (+)	4.1* (+)	4.1** (+)	3.9** (+)	3.8** (+)	3.1
	Ph.D.	72	3.8** (+)	3.6* (+)	3.6** (+)	3.4** (+)	3.1** (+)	3.1
New TA	experienced	220	3.9	3.7	3.7	3.5	3.3	3.1
	new	93	3.8	3.8	3.7	3.5	3.3	3.3
Training	not in person	151	3.8	3.7	3.7	3.3	3.2	3.2
	in person	162	3.9	3.8	3.8	3.6	3.4	3.1
Training async	no asynch training	225	3.8	3.7	3.7	3.6	3.4	3.1
	received async training	88	3.9	3.8	3.7	3.4	3.2	3.1
Training in person + async	did not receive both	274	3.8	3.7	3.7	3.5	3.4	3.2
	received both	39	4.1	3.9	3.8	3.4	3.2	3.0
Training no training	received training	211	n/a	n/a	n/a	n/a	n/a	3.1
	did not receive training	102	n/a	n/a	n/a	n/a	n/a	3.2
is baseline	intervention	190	3.9	3.8	3.8	3.6	3.5	3.2
	baseline	123	3.8	3.7	3.7	3.4	3.1	3.1

Table 21. Mean values for TA-TQ questions for every tested dimension; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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C.2 Full Student Results

		Univ. 1		Univ. 2		Univ. 3		Univ. 4		Univ. 5	
factor	value	mean	N	mean	N	mean	N	mean	N	mean	N
Gender	M	4.1	244	4.4	55	4.3	426	4.2	57	4.4	393
	Oth.					4.1	22				
	W	4.2	102	4.6	29	4.2	215	4.5	24	4.4	170
Race	Black	4.2	112			4.0	40	4.3** (+++)	13	4.4** (+)	40
	Arab/ME					4.3	17			4.2** (+)	26
	White	4.2	28			4.3	334	4.4** (+++)	32	4.4** (+)	166
	E. Asian	4.2	38	4.3	14	4.3	69			4.0** (+)	23
	Oth. Asian					4.1	12			4.4** (+)	21
	Other	4.6	11			4.3	43			3.8** (+)	17
	S. Asian	4.1	111	4.5	38	4.1	54	4.4** (+++)	18	4.5** (+)	150
	SE. Asian	3.8	29	4.2	10	3.8	22			4.4** (+)	77
	Multi	3.9	19			4.1	58	3.2** (+++)	11	4.4** (+)	43
Black intersectional	Black M	4.2	69			4.3	21			4.4	27
	Black W	4.2	37			4.0	14			4.7	12
White intersectional	White M	4.2	17			4.3	201	4.3	23	4.4* (+)	124
	White W					4.3	110			4.1* (+)	32
E. Asian intersectional	E. Asian M	4.2	27			4.4	34			3.8	12
	E. Asian W					4.1	28				
Other intersectional	Other M					4.4	25				
	Other W					4.2	13				
S. Asian intersectional	S. Asian M	4.1	76	4.6	24	4.1	44	4.5	11	4.4	91
	S. Asian W	4.1	32	4.4	13					4.6	58
SE. Asian intersectional	SE. Asian M	3.8	22			3.8	17			4.6	44
	SE. Asian W									4.4	26
Multi intersectional	Multi M	4.0	13			4.2	34			4.3	32
	Multi W					4.1	21			4.5	11
Eth.	Other Hisp.	4.5	11			4.4	40			4.2	60
	Cuban									4.5	12
	Chicanx	4.2	13			4.3	111				
	Puerto Rican									4.2	12
	Multi					4.1	13			4.4	12
	Not H/L/S	4.1	308	4.5	60	4.2	461	4.2	72	4.4	442
Dis.	None	4.2** (+++)	320	4.4	72	4.3*** (+++)	499	4.4*** (+++)	60	4.4	489
	Yes	3.8** (+++)	40			3.9*** (+++)	119	3.5*** (+++)	23	4.1	64
Course	CS0			4.3	39	4.6*** (+)	53				
	CS1	4.0	97	4.7	10	4.2*** (+)	296	4.5	20	4.3	207
	CS2	4.2	214	4.4	44	4.3*** (+)	165			4.4	251
	CS3	4.2	53			3.9*** (+)	93			4.4	140
	Discrete 1	4.3	10			4.4*** (+)	71	4.1	36		
	Other					4.3*** (+)	39	4.4	19		
Term	SP2023			4.5	15			3.9* (++)	31		
	FA2023	4.1	168	4.3	19	4.4*** (+)	356	4.2* (++)	35	4.3	107
	SP2024	4.2	103	4.7	19	4.2*** (+)	232			4.4	164
	FA2024	4.1	103	4.3	40	4.0*** (+)	129	4.6* (++)	17	4.4	327

Table 22. Mean values for S-OQ1 for every tested dimension; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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factor	value	Univ. 1 mean	Univ. 1 N	Univ. 2 mean	Univ. 2 N	Univ. 3 mean	Univ. 3 N	Univ. 4 mean	Univ. 4 N	Univ. 5 mean	Univ. 5 N
Gender	M	4.0	243	4.2	55	4.0	425	3.9	57	4.1	393
	Oth.					3.7	22				
	W	3.9	101	4.3	29	3.9	215	4.2	24	4.2	170
Race	Black	4.0	111			3.8	40	3.8** (++)	13	4.2	40
	Arab/ME					4.0	17			3.8	26
	White	4.1	28			4.0	334	4.1** (++)	32	4.2	166
	E. Asian	3.9	38	3.9	14	4.0	69			3.9	23
	Oth. Asian					3.8	12			4.1	21
	Other	4.6	11			4.0	42			3.8	17
	S. Asian	3.9	111	4.4	38	3.9	54	4.2** (++)	18	4.2	150
	SE. Asian	3.8	29	4.1	10	3.5	22			4.1	77
	Multi	3.8	18			3.8	58	2.9** (++)	11	4.1	43
Black intersectional	Black M	4.1	69			4.0	21			4.1	27
	Black W	4.0	36			3.8	14			4.5	12
White intersectional	White M	4.1	17			4.1	201	4.0	23	4.2	124
	White W					3.9	110			4.1	32
E. Asian intersectional	E. Asian M	4.0	27			4.2	34			3.6	12
	E. Asian W					3.8	28				
Other intersectional	Other M					4.1	24				
	Other W					4.0	13				
S. Asian intersectional	S. Asian M	4.0	76	4.5	24	3.9	44	4.4	11	4.1	91
	S. Asian W	3.8	32	4.2	13					4.3	58
SE. Asian intersectional	SE. Asian M	3.6	22			3.6	17			4.2	44
	SE. Asian W									4.2	26
Multi intersectional	Multi M	3.8	12			3.9	34			4.1	32
	Multi W					3.8	21			4.2	11
Eth.	Other Hisp.	4.5	11			4.0	40			4.2	60
	Cuban									4.5	12
	Chicanx	4.0	13			4.0	110				
	Puerto Rican									4.2	12
	Multi					3.8	13			4.0	12
	Not H/L/S	3.9	307	4.2	60	3.9	461	3.9	72	4.1	442
Dis.	None	4.0** (+++)	318	4.2	72	4.0*** (+++)	498	4.0** (+++)	60	4.2	489
	Yes	3.5** (+++)	40			3.7*** (+++)	119	3.3** (+++)	23	4.0	64
Course	CS0			4.1	39	4.3*** (+)	53				
	CS1	3.8	97	4.4	10	3.9*** (+)	296	4.2	20	4.0	207
	CS2	4.0	212	4.2	44	4.0*** (+)	165			4.2	251
	CS3	4.0	53			3.7*** (+)	93			4.1	140
	Discrete 1	3.9	10			4.0*** (+)	71	3.8	36		
	Other					4.0*** (+)	38	4.2	19		
Term	SP2023			4.3	15			3.6* (++)	31		
	FA2023	3.9	168	4.0	19	4.1*** (+)	355	3.9* (++)	35	4.1	107
	SP2024	4.0	103	4.4	19	3.9*** (+)	232			4.2	164
	FA2024	3.9	101	4.1	40	3.8*** (+)	129	4.5* (++)	17	4.1	327

Table 23. Mean values for S-OQ2 for every tested dimension; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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factor	value	Univ. 1 mean	Univ. 1 N	Univ. 2 mean	Univ. 2 N	Univ. 3 mean	Univ. 3 N	Univ. 4 mean	Univ. 4 N	Univ. 5 mean	Univ. 5 N
Gender	M	3.9	243	4.0	55	4.0	426	3.8	57	4.1	393
	Oth.					3.9	22				
	W	3.9	102	4.2	29	3.9	214	4.2	24	4.2	170
Race	Black	3.9	111			3.6	40	3.8** (++)	13	4.2* (+)	40
	Arab/ME					3.8	17			4.0* (+)	26
	White	4.0	28			4.0	334	4.0** (++)	32	4.1* (+)	166
	E. Asian	3.9	38	3.9	14	4.0	68			3.7* (+)	23
	Oth. Asian					3.9	12			4.1* (+)	21
	Other	4.5	11			4.1	43			3.8* (+)	17
	S. Asian	3.9	111	4.4	38	3.9	54	4.2** (++)	18	4.3* (+)	150
	SE. Asian	3.8	29	4.1	10	3.5	22			4.2* (+)	77
	Multi	3.7	19			3.7	58	2.8** (++)	11	4.1* (+)	43
Black intersectional	Black M	4.0	68			3.8	21			4.1	27
	Black W	3.9	37			3.4	14			4.5	12
White intersectional	White M	4.1	17			4.0	201	3.8	23	4.1	124
	White W					4.0	110			3.9	32
E. Asian intersectional	E. Asian M	4.0	27			4.2	34			3.4	12
	E. Asian W					3.9	27				
Other intersectional	Other M					4.1	25				
	Other W					3.9	13				
S. Asian intersectional	S. Asian M	4.0	76	4.2	24	3.9	44	4.3	11	4.2	91
	S. Asian W	3.7	32	4.5	13					4.4	58
SE. Asian intersectional	SE. Asian M	3.6	22			3.5	17			4.3	44
	SE. Asian W									4.3	26
Multi intersectional	Multi M	3.8	13			3.7	34			4.2	32
	Multi W					3.7	21			4.1	11
Eth.	Other Hisp.	4.4	11			3.9	40			4.0	60
	Cuban									4.3	12
	Chicanx	3.9	13			4.0	111				
	Puerto Rican									4.0	12
	Multi					4.0	13			4.1	12
	Not H/L/S	3.9	307	4.1	60	3.9	460	3.8	72	4.1	442
Dis.	None	3.9** (+++)	319	4.1	72	4.0*** (+++)	498	4.0** (+++)	60	4.2	489
	Yes	3.6** (+++)	40			3.7*** (+++)	119	3.2** (+++)	23	3.9	64
Course	CS0			3.9	39	4.1** (+)	53				
	CS1	3.7	96	4.2	10	3.9** (+)	296	4.2	20	4.0	207
	CS2	4.0	214	4.1	44	4.0** (+)	165			4.2	251
	CS3	4.0	53			3.6** (+)	93			4.2	140
	Discrete 1	3.9	10			4.0** (+)	71	3.6	36		
	Other					4.1** (+)	38	4.2	19		
Term	SP2023			4.0	15			3.7* (++)	31		
	FA2023	3.8	167	4.0	19	4.0** (-)	355	3.7* (++)	35	4.1	107
	SP2024	4.1	103	4.4	19	3.9** (-)	232			4.2	164
	FA2024	3.8	103	3.9	40	3.8** (-)	129	4.5* (++)	17	4.1	327

Table 24. Mean values for S-OQ3 for every tested dimension; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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		Univ. 1		Univ. 2		Univ. 3		Univ. 4		Univ. 5	
factor	value	%	N	%	N	%	N	%	N	%	N
Gender	M	26.8	912	18.3	300	59.1*** (+)	722	33.3	171	47.1*** (-)	835
	Oth.	24.1	29	0.0	13	71.0*** (+)	31			30.0*** (-)	20
	W	25.9	402	19.0	153	72.9*** (+)	295	41.4	58	56.5*** (-)	301
Race	Black	32.3*** (+)	356	8.3* (+)	12	83.3*** (+)	48	46.4	28	47.6	84
	Arab/ME	30.4*** (+)	23			60.7*** (+)	28			54.2	48
	White	17.9*** (+)	156	14.5* (+)	55	60.7*** (+)	552	28.3	113	43.5	382
	E. Asian	23.9*** (+)	163	16.5* (+)	85	74.2*** (+)	93	27.3	11	52.3	44
	AIAN					45.5*** (+)	11				
	Oth. Asian					63.2*** (+)	19			52.5	40
	Other	36.7*** (+)	30	16.7* (+)	12	79.6*** (+)	54			41.5	41
	S. Asian	29.1*** (+)	382	25.3* (+)	150	78.3*** (+)	69	42.9	42	54.3	276
	SE. Asian	21.0*** (+)	138	11.1* (+)	90	52.4*** (+)	42			55.8	138
	Multi	20.9*** (+)	91	12.5* (+)	48	52.3*** (+)	111	45.8	24	47.8	90
Black intersectional	Black M	31.1	222			75.0	28	41.2	17	47.4	57
	Black W	33.1	118			93.3	15	50.0	10	50.0	24
White intersectional	White M	17.0	100	13.3	30	54.4*** (+)	371	27.7	83	43.1	288
	White Oth.	18.2	11			80.0*** (+)	10			41.7	12
	White W	21.1	38	20.0	20	73.3*** (+)	150	22.7	22	47.1	68
E. Asian intersectional	E. Asian M	23.7	114	17.0	53	64.2*** (+++)	53			50.0	24
	E. Asian W	21.1	38	17.9	28	90.3*** (+++)	31			47.1	17
Other intersectional	Other M					73.5	34			39.1	23
	Other W					92.9	14			50.0	16
S. Asian intersectional	S. Asian M	30.3	251	26.7	90	81.5	54	35.5* (+++)	31	47.9*** (+++)	190
	S. Asian W	28.1	114	24.1	54	75.0	12	70.0* (+++)	10	69.9*** (+++)	83
SE. Asian intersectional	SE. Asian M	23.4	94	14.3	63	58.6	29			48.4* (+++)	91
	SE. Asian W	17.5	40	4.8	21	38.5	13			65.0* (+++)	40
Multi intersectional	Multi M	21.7	60	11.1	27	49.3	69			53.3	60
	Multi W	8.3	24	20.0	15	61.8	34			40.7	27
Eth.	Other Hisp.	19.3	57	0.0	11	69.0	58	33.3	15	48.4	124
	Cuban									50.0	24
	Chicanx	20.0	65	22.5	40	64.2	173	28.6	28	28.0	25
	Puerto Rican	40.0	20			63.6	11			42.9	28
	Multi	35.7	14			68.4	19			60.0	20
	Not H/L/S	26.9	1153	16.8	357	61.9	746	38.9	185	50.0	884
Dis.	None	26.7	1209	17.4	414	62.3	802	35.5	169	49.6** (+++)	985
	Yes	23.8	168	11.5	61	57.8	206	35.4	65	40.8** (+++)	157
Course	CS0			20.7	213	63.3*** (+)	90	14.3* (+)	21		
	CS1	31.6*** (+)	320	16.7	78	75.6*** (+)	426	32.9* (+)	82	39.7*** (+)	536
	CS2	26.1*** (+)	876	21.3	235	63.7*** (+)	273	41.9* (+)	31	58.9*** (+)	440
	CS3	27.6*** (+)	239			67.1*** (+)	143			57.2*** (+)	250
	Discrete 1	73.3*** (+)	15			44.8*** (+)	163	45.4* (+)	97		
	Other					57.3*** (+)	75	46.3* (+)	41		
Term	SP2023			32.8** (+)	61			35.7	115		
	FA2023	32.5*** (-)	542	22.1** (+)	113	62.9	615	45.4	97	45.7** (-)	247
	SP2024	27.0*** (-)	430	23.1** (+)	91	69.4	359	31.6	19	46.7** (-)	368
	FA2024	24.1*** (-)	478	15.7** (+)	261	65.8	196	39.5	43	54.0** (-)	611

Table 25. Percent students attending office hours; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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factor	value	Univ. 1 %	Univ. 1 N	Univ. 2 %	Univ. 2 N	Univ. 3 %	Univ. 3 N	Univ. 4 %	Univ. 4 N	Univ. 5 %	Univ. 5 N
Gender	M	70.1*** (+)	668	68.6*** (+)	245	69.2	295	73.7	114	69.7	442
	Oth.	59.1*** (+)	22	38.5*** (+)	13					78.6	14
	W	52.7*** (+)	298	50.8*** (+)	124	61.3	80	76.5	34	64.9	131
Race	Black	54.8*** (+)	241	36.4*** (+)	11			60.0	15	59.1	44
	Arab/ME	62.5*** (+)	16			90.9	11			54.5	22
	White	74.2*** (+)	128	57.4*** (+)	47	70.0	217	79.0	81	70.4	216
	E. Asian	56.5*** (+)	124	62.0*** (+)	71	66.7	24			66.7	21
	Oth. Asian									52.6	19
	Other	63.2*** (+)	19	20.0*** (+)	10	54.5	11			66.7	24
	S. Asian	73.8*** (+)	271	75.0*** (+)	112	40.0	15	62.5	24	75.4	126
	SE. Asian	61.5*** (+)	109	65.0*** (+)	80	85.0	20			67.2	61
	Multi	66.7*** (+)	72	54.8*** (+)	42	66.0	53	61.5	13	76.6	47
Black intersectional	Black M	58.8	153					70.0	10	60.0	30
	Black W	46.8	79							50.0	12
White intersectional	White M	78.3	83	69.2	26	72.8	169	80.0	60	71.3	164
	White W	70.0	30	37.5	16	57.5	40	88.2	17	61.1	36
E. Asian intersectional	E. Asian M	66.7*** (+++)	87	68.2	44	68.4	19			66.7	12
	E. Asian W	33.3*** (+++)	30	56.5	23						
S. Asian intersectional	S. Asian M	78.9* (+++)	175	81.8* (+++)	66	40.0	10	65.0	20	76.8	99
	S. Asian W	64.6* (+++)	82	61.0* (+++)	41					76.0	25
SE. Asian intersectional	SE. Asian M	62.5	72	64.8	54	83.3	12			68.1	47
	SE. Asian W	54.5	33	65.0	20					64.3	14
Multi intersectional	Multi M	72.3	47	79.2*** (+++)	24	68.6	35			78.6	28
	Multi W	59.1	22	16.7*** (+++)	12	69.2	13			75.0	16
Eth.	Other Hisp.	65.2	46	36.4** (+)	11	55.6	18	70.0	10	65.6** (+)	64
	Cuban									83.3** (+)	12
	Chicanx	51.9	52	38.7** (+)	31	61.3	62	85.0	20	61.1** (+)	18
	Puerto Rican	58.3	12							37.5** (+)	16
	Not H/L/S	65.5	843	64.3** (+)	297	68.3	284	71.7	113	71.7** (+)	442
Dis.	None	65.7* (+++)	886	64.9*** (+++)	342	69.5	302	74.3	109	70.2	496
	Yes	54.7* (+++)	128	42.6*** (+++)	54	60.9	87	69.0	42	66.7	93
Course	CS0			56.8	169	60.6	33	72.2	18		
	CS1	63.0	219	60.0	65	73.1	104	70.9	55	74.9*** (+)	323
	CS2	65.2	647	67.0	185	66.7	99	66.7	18	65.2*** (+)	181
	CS3	61.8	173			72.3	47			55.1*** (+)	107
	Discrete 1					60.0	90	69.8	53		
	Other					65.6	32	86.4	22		
Term	SP2023			65.9	41			68.9	74		
	FA2023	67.8	366	58.0	88	70.2	228	71.7	53	79.9*** (+)	134
	SP2024	64.0	314	57.1	70	61.8	110	69.2	13	73.5*** (+)	196
	FA2024	60.3	363	64.1	220	64.2	67	84.6	26	59.8*** (+)	281

Table 26. Percent students selecting S-NOH1, "I did not need help"; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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		Univ. 1		Univ. 2		Univ. 3		Univ. 4		Univ. 5	
factor	value	%	N	%	N	%	N	%	N	%	N
Gender	M	35.2	668	34.3	245	35.9	295	21.1	114	32.6	442
	Oth.	36.4	22	46.2	13					28.6	14
	W	41.9	298	36.3	124	33.8	80	35.3	34	29.8	131
Race	Black	41.1* (+)	241	27.3	11			40.0	15	45.5	44
	Arab/ME	37.5* (+)	16			9.1	11			40.9	22
	White	32.0* (+)	128	34.0	47	38.7	217	19.8	81	31.9	216
	E. Asian	34.7* (+)	124	39.4	71	37.5	24			42.9	21
	Oth. Asian									36.8	19
	Other	57.9* (+)	19	60.0	10	54.5	11			37.5	24
	S. Asian	32.5* (+)	271	35.7	112	26.7	15	12.5	24	20.6	126
	SE. Asian	45.0* (+)	109	35.0	80	25.0	20			37.7	61
	Multi	36.1* (+)	72	26.2	42	26.4	53	38.5	13	23.4	47
Black intersectional	Black M	40.5	153					40.0	10	53.3	30
	Black W	43.0	79							33.3	12
White intersectional	White M	25.3	83	30.8	26	38.5	169	13.3** (+++)	60	31.1	164
	White W	40.0	30	43.8	16	40.0	40	41.2** (+++)	17	38.9	36
E. Asian intersectional	E. Asian M	28.7* (+++)	87	38.6	44	36.8	19			50.0	12
	E. Asian W	50.0* (+++)	30	34.8	23						
S. Asian intersectional	S. Asian M	33.7	175	34.8	66	20.0	10	15.0	20	19.2	99
	S. Asian W	31.7	82	41.5	41					24.0	25
SE. Asian intersectional	SE. Asian M	41.7	72	33.3	54	33.3	12			38.3	47
	SE. Asian W	54.5	33	35.0	20					35.7	14
Multi intersectional	Multi M	36.2	47	25.0	24	34.3	35			25.0	28
	Multi W	31.8	22	25.0	12	15.4	13			18.8	16
Eth.	Other Hisp.	45.7	46	18.2	11	61.1	18	30.0	10	28.1	64
	Cuban									25.0	12
	Chicanx	50.0	52	45.2	31	40.3	62	35.0	20	33.3	18
	Puerto Rican	33.3	12							56.2	16
	Not H/L/S	36.8	843	34.0	297	33.5	284	21.2	113	30.3	442
Dis.	None	37.5	886	34.5	342	34.1	302	22.9	109	31.2	496
	Yes	39.1	128	40.7	54	35.6	87	26.2	42	32.3	93
Course	CS0			32.5	169	42.4** (+)	33	22.2	18		
	CS1	32.9	219	33.8	65	23.1** (+)	104	12.7	55	29.7	323
	CS2	37.1	647	37.3	185	30.3** (+)	99	11.1	18	32.0	181
	CS3	41.6	173			40.4** (+)	47			33.6	107
	Discrete 1					48.9** (+)	90	35.8	53		
	Other					28.1** (+)	32	31.8	22		
Term	SP2023			24.4** (+)	41			29.7	74		
	FA2023	30.3*** (+)	366	34.1** (+)	88	28.9* (+)	228	17.0	53	26.1** (+)	134
	SP2024	37.9*** (+)	314	52.9** (+)	70	40.9* (+)	110	7.7	13	26.0** (+)	196
	FA2024	43.0*** (+)	363	31.4** (+)	220	43.3* (+)	67	26.9	26	37.0** (+)	281

Table 27. Percent students selecting S-NOH2, "The office hour times were inconvenient for my schedule"; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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factor	value	Univ. 1 %	Univ. 1 N	Univ. 2 %	Univ. 2 N	Univ. 3 %	Univ. 3 N	Univ. 4 %	Univ. 4 N	Univ. 5 %	Univ. 5 N
Gender	M	6.6** (-)	668	4.1	245	4.1	295	7.0	114	2.3	442
	Oth.	22.7** (-)	22	7.7	13					7.1	14
	W	9.4** (-)	298	8.1	124	10.0	80	8.8	34	4.6	131
Race	Black	7.9** (+)	241	18.2	11			6.7	15	0.0	44
	Arab/ME	12.5** (+)	16			0.0	11			0.0	22
	White	10.2** (+)	128	6.4	47	4.6	217	7.4	81	4.2	216
	E. Asian	11.3** (+)	124	2.8	71	0.0	24			0.0	21
	Oth. Asian									0.0	19
	Other	0.0** (+)	19	10.0	10	9.1	11			4.2	24
	S. Asian	3.7** (+)	271	1.8	112	20.0	15	8.3	24	3.2	126
	SE. Asian	6.4** (+)	109	6.2	80	0.0	20			3.3	61
	Multi	13.9** (+)	72	7.1	42	3.8	53	15.4	13	0.0	47
Black intersectional	Black M	7.2	153					10.0	10	0.0	30
	Black W	10.1	79							0.0	12
White intersectional	White M	7.2	83	3.8	26	3.0** (+)	169	6.7	60	3.0	164
	White W	10.0	30	12.5	16	12.5** (+)	40	5.9	17	8.3	36
E. Asian intersectional	E. Asian M	11.5	87	4.5	44	0.0	19			0.0	12
	E. Asian W	6.7	30	0.0	23						
S. Asian intersectional	S. Asian M	3.4	175	1.5	66	30.0	10	5.0	20	2.0	99
	S. Asian W	4.9	82	2.4	41					8.0	25
SE. Asian intersectional	SE. Asian M	4.2	72	3.7	54	0.0	12			4.3	47
	SE. Asian W	12.1	33	10.0	20					0.0	14
Multi intersectional	Multi M	8.5	47	0.0* (+++)	24	2.9	35			0.0	28
	Multi W	27.3	22	16.7* (+++)	12	7.7	13			0.0	16
Eth.	Other Hisp.	19.6*** (+)	46	18.2** (+)	11	0.0	18	0.0	10	4.7*** (+)	64
	Cuban									25.0*** (+)	12
	Chicanx	9.6*** (+)	52	16.1** (+)	31	4.8	62	0.0	20	0.0*** (+)	18
	Puerto Rican	33.3*** (+)	12							6.2*** (+)	16
	Not H/L/S	6.8*** (+)	843	4.7** (+)	297	5.6	284	8.0	113	2.0*** (+)	442
Dis.	None	7.0	886	4.1** (+++)	342	4.6	302	3.7** (+++)	109	2.4	496
	Yes	11.7	128	13.0** (+++)	54	6.9	87	16.7** (+++)	42	5.4	93
Course	CS0			7.7	169	9.1* (+)	33	5.6	18		
	CS1	11.4** (-)	219	7.7	65	1.0* (+)	104	12.7	55	3.7	323
	CS2	6.0** (-)	647	4.3	185	10.1* (+)	99	11.1	18	1.7	181
	CS3	10.4** (-)	173			8.5* (+)	47			1.9	107
	Discrete 1					2.2* (+)	90	5.7	53		
	Other					3.1* (+)	32	0.0	22		
Term	SP2023			4.9	41			8.1	74		
	FA2023	7.9	366	9.1	88	5.7	228	5.7	53	4.5	134
	SP2024	7.6	314	7.1	70	6.4	110	15.4	13	2.6	196
	FA2024	8.3	363	5.0	220	1.5	67	7.7	26	2.1	281

Table 28. Percent students selecting S-NOH3, "I did not feel comfortable with any of the TAs for this course"; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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		Univ. 1		Univ. 2		Univ. 3		Univ. 4		Univ. 5	
factor	value	%	N	%	N	%	N	%	N	%	N
Gender	M	7.6** (-)	668	2.4*** (+)	245	4.1	295	12.3	114	7.9	442
	Oth.	0.0** (-)	22	15.4*** (+)	13					7.1	14
	W	12.8** (-)	298	11.3*** (+)	124	8.8	80	20.6	34	5.3	131
Race	Black	12.9** (+)	241	45.5*** (++)	11			33.3	15	6.8	44
	Arab/ME	18.8** (+)	16			0.0	11			13.6	22
	White	7.0** (+)	128	4.3*** (++)	47	5.1	217	13.6	81	8.8	216
	E. Asian	4.0** (+)	124	1.4*** (++)	71	12.5	24			4.8	21
	Oth. Asian									10.5	19
	Other	5.3** (+)	19	30.0*** (++)	10	0.0	11			4.2	24
	S. Asian	9.2** (+)	271	5.4*** (++)	112	0.0	15	16.7	24	10.3	126
	SE. Asian	5.5** (+)	109	1.2*** (++)	80	5.0	20			0.0	61
	Multi	15.3** (+)	72	9.5*** (++)	42	1.9	53	7.7	13	0.0	47
Black intersectional	Black M	10.5	153					30.0	10	6.7	30
	Black W	17.7	79							8.3	12
White intersectional	White M	4.8	83	3.8	26	3.6	169	8.3	60	9.8	164
	White W	13.3	30	6.2	16	10.0	40	17.6	17	5.6	36
E. Asian intersectional	E. Asian M	4.6	87	0.0	44	10.5	19			8.3	12
	E. Asian W	0.0	30	4.3	23						
S. Asian intersectional	S. Asian M	8.0	175	4.5	66	0.0	10	10.0	20	9.1	99
	S. Asian W	12.2	82	7.3	41					12.0	25
SE. Asian intersectional	SE. Asian M	4.2	72	0.0	54	0.0	12			0.0	47
	SE. Asian W	9.1	33	5.0	20					0.0	14
Multi intersectional	Multi M	12.8	47	0.0** (+++)	24	2.9	35			0.0	28
	Multi W	22.7	22	25.0** (+++)	12	0.0	13			0.0	16
Eth.	Other Hisp.	15.2** (+)	46	18.2	11	0.0	18	10.0	10	10.9** (+)	64
	Cuban									33.3** (+)	12
	Chicanx	3.8** (+)	52	0.0	31	6.5	62	10.0	20	11.1** (+)	18
	Puerto Rican	33.3** (+)	12							0.0** (+)	16
	Not H/L/S	8.9** (+)	843	6.4	297	4.9	284	16.8	113	6.8** (+)	442
Dis.	None	9.3	886	5.3	342	3.3* (+++)	302	12.8	109	7.1	496
	Yes	7.8	128	11.1	54	9.2* (+++)	87	23.8	42	8.6	93
Course	CS0			8.3	169	0.0	33	16.7	18		
	CS1	17.4*** (+)	219	7.7	65	3.8	104	18.2	55	4.3** (+)	323
	CS2	6.2*** (+)	647	3.8	185	9.1	99	22.2	18	9.9** (+)	181
	CS3	11.6*** (+)	173			4.3	47			12.1** (+)	107
	Discrete 1					3.3	90	13.2	53		
	Other					9.4	32	4.5	22		
Term	SP2023			0.0	41			17.6	74		
	FA2023	12.0	366	10.2	88	6.1	228	15.1	53	8.2	134
	SP2024	7.6	314	5.7	70	6.4	110	23.1	13	4.1	196
	FA2024	8.3	363	5.9	220	0.0	67	3.8	26	9.3	281

Table 29. Percent students selecting S-NOH4, "I did not believe the TAs would be able to help me with my needs for this course"; * indicates significant at $p < 0.1$, ** at $p < 0.05$, *** at $p < 0.01$; all p values indicated calculated with Benjamini-Hochberg corrections; effect sizes are shown in parentheses, - indicates insignificant, + small, ++ medium, +++ large

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